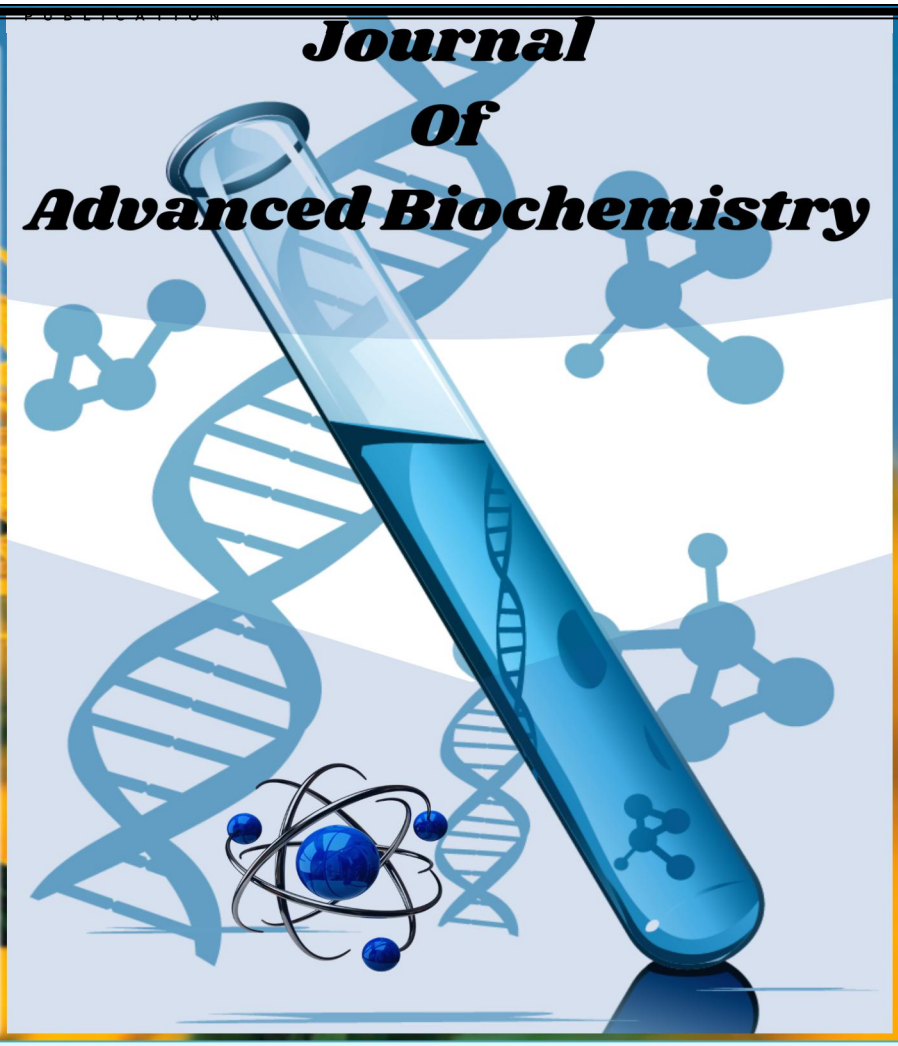
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PUBLICATION

Journal Of Advanced Biochemistry



LION GROUPS FOR ZOOLOGISTS

Group properties:

- ♥ Separate group for each subject
- ♥ Video lectures
- ♥ Ideal assignments, GDBs
- ♥ Final term and midterm preparations

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Final term 2022

EFFORT BY: MAAH WARRIORS

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ASSESSMENT BIO202

mid term

Week no.1

TOPIC NO.1

1. Crystalline derivatives of monosaccharide, formed from pheny hydrazine are known as…
   a. Cake
   b. **osazone**
   c. sugar
   d. glucose
2. Simplest carbohydrate is….
   a. Glucose
   b. **glyceraldehyde**
   c. glycerine
   d. sucrose
3. In Benedict`stest the…..are produced which form red precipitate
   a. Ferric ions
   b. cupric ions
   c. **cuprous ion**
   d. ferrous ions
4. Monosaccharides can be subdivided further depending upon…
   a. Aldehyde groups
   b. number of carbon atoms
   c. **number of carbon atoms and aldehyde or ketone groups**
   d. none of given
5. If the carbonyl group is at an end of carbon chain (that is in a….) the monosaccharides is an aldose.
   a. **Aldehyde group**
   b. carbonyl group
   c. carboxyl group
   d. ketone group

TOPIC NO.2

6. The terminal methyl and…..carbon of glucose, galactose, or mannose-forms the corresponding aldaric acid.
   a. Ketone
   b. **carbonyl**
   c. carboxyl
   d. aldehyde
7. Carbohydrates are polyhydroxy….or substances that yield such compounds on hydrolysis.
   a. Ketones
   b. **aldehydes or ketones**
   c. aldehydes
   d. all of given
8. …..is commonly found in fruit juices, personal lubricants, and wine and is used to create tangy taste in food.
   a. Glucose
   b. **glucose delta-lactone**
   c. salt
   d. honey
9. ….is semi-quantitative test to detect presence of glucose in a solution
   a. **Benedicts test**
   b. molisch`s test
   c. ninhydrin test
   d. none of given
10. In the open chain form of carbohydrates one of carbon atoms is double-bonded to an oxygen atom to form a….
    a. **Carbonyl group**
    b. ketone group
    c. carboxtl group
    d. aldehyde group

TOPIC NO.3

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11. Both glycogen and cellulose are made up of glucose subunits, however while glycogen is…., cellulose exists in the form of…

a. Disaccharide, polysaccharide
b. polysaccharide, disaccharide
c. linear chains, branched
d. **branched, linear chain**

TOPIC NO.6

12. The designation of a sugar isomer as the D form or as the L form is determined by its spatial relationship to parent compound of carbohydrates which is…..

a. Glucose
b. glycogen
c. **glyceraldehyde**
d. glycerol

TOPIC NO.7

13. With reference to carbohydrate structure, which type of isomers has nearly identical chemical properties but differ in their interaction with plane-polarized light?

a. Anomers
b. aldose-ketose isomerism
c. **enantiomers**
d. epimers

TOPIC NO.9

14. Isomeric forms of monosaccharide that differ only in their configuration about hemiacetal or hemiketal carbon atom are called….

a. Epimers
b. stereoisomers
c. **anormes**
d. enantiomers

TOPIC NO.11

15. Among six membered cyclic carbohydrates, the….is much more stable than the aldofuranose ring.

a. **Aldopyranose ring**
b. ketopyranose ring
c. ketofuranose ring
d. aldopentose ring

TOPIC NO.12

16. Sugars are classified on the basis of number of carbon atoms present. An example of a tetrose is:

a. **erythrose**
b. glucose
c. ribose
d. glyceraldehyde

17. All of following statements about glucose are correct except:

a. It is the precursor for all other carbohydrates in body
b. it is chief physiological sugar present in normal blood at a fairly constant level
c. it is present in fruit juices
d. **it is mostly exists in L form**

18. Glucose is excreted in urine in poorly controlled diabetes mellitus as a result of hyperglycemia. This medical condition is known as:

a. glyceraldehyde 3 dehydrogenase deficiency
b. **glucosuria**
c. Hypoglycemia
d. hyperinsulinism

19. Glucose and galactose are epimers which differ in configuration around carbon no. …..

a. 1
b. 2
c. 6
d. 4

TOPIC NO.13

20. The oxidation of carbonyl carbon in monosaccharide yield products known as:

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a. **Aldonic acids** c. uronic acids
b. furfurals d. aldoses

21. ….is a levorotatory sugar and also known as levulose.
a. Galactose c. mannose
b. **fructose** d. glucose

22. Different sugars are utilized in various ways in human body and hence occur in varying amounts in body fluids, which fluid is rich in fructose?
a. **Seminal fluid** c. blood
b. cerebrospinal fluid d. tissue fluid

23. The oxidation of monosaccharide results in production of either aldonic acids or uronic acids depending on the carbon atoms oxidized. Both aldonic and uronic acids form stable intramolecular ester that are known as….
a. **Lactones** c. furfurals
b. carboxylates d. aldoses

24. The condition in which the body fails to metabolize galactose is known as:
a. Hypoglycemia c. diabetes mellitus
b. glucosuria d. **galactosemia**

TOPIC NO.14

25. glucone delta-lactone (GDL) is a type of lactone, commonly found in honey, fruit juices, personal lubricants, and wine. It hydrolysis in water to form…..which has a….. PH
a. glucurenic acid, acidic c. glucose, neutral
b. **gluconic acid, acidic** d. glyceraldehyde, neutral

26. glucuronide is a soluble, non-toxic compound that is excreted in urine. It is produced when….forms conjugates with toxic substances including drugs, hormones and bilirubin.
a. Gluconic acid c. glucose
b. glyceraldehyde d. **glucuronic acid**

TOPIC NO.15

27. Carbohydrates are detected in laboratory based upon certain properties. Benedict`s test is routinely used to check for presence of….
a. Aldopentoses c. **reducing sugar**
b. ketohexoses d. polysaccharides

TOPIC NO.16

28. In synthesis and metabolism of carbohydrates, the intermediates are very often not sugars themselves but their….derivatives.
a. carboxylates c. **phosphorylated**
b. oxidized d. methylated

TOPIC NO.17

29. Amino sugars are formed when hydroxyl group in sugar is replaced with an amino group. An amino sugar that is a constituent of chondroitin is….
a. Sialic acid c. D-mannosamine
b. **D-galactosamine** d. D-glucoasmine

30. Many antibiotics, like erythromycin, contain sugar derivatives such as…..those are important for their antibiotic activity….

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a. Carboxylated sugar
b. phosphorylated sugars
c. **amino sugars**
d. oxidizing sugars

Week no.2

TOPIC NO.18

31. When the oxygen from the –OH group in a monosaccharide is removed, the resulting compound formed is known as….

a. Amino sugar
b. phosphor sugar
c. **deoxy sugar**
d. carboxyl sugar

32. The substitution of hydrogen for hydroxyl group at C-6 of L-mannose produces….

a. L-glucose
b. D-mannose
c. L-mannosamine
d. **L-rhamnose**

TOPIC NO.20

33. Osazones are obtained by adding a mixture of phenyl hydrazine hydrochloride and sodium acetate to a sugar solution and heating in water bath for 45 mins. Which two carbon atoms in the sugar molecule are involved in osazone formation?

a. **C1 and C2**
b. C1 and C6
c. C1 and C4
d. C3 and C5

34. Maltosazone is the osazone formed from maltose. The characteristic feature of this osazone is its shape which resembles a….

a. **Sunflower**
b. needle
c. pin-cushion
d. powder-puff

TOPIC NO.21

35. Upon reaction with strong acids pentoses produce…..while hexoses produce….

a. Deoxy sugars, amino sugars
b. **furfural, hydroxyl methyl furfural**
c. hydroxyl methyl furfural, furfural
d. aldehydes, ketone

36. The formation of furfural products and their condensation with organic phenols to give characteristics colored compounds forms the basis of biochemical tests used for detection of carbohydrates. An example of such a test is….

a. **Molisch`s test**
b. ninhydrine test
c. benedict`s test
d. grease spot test

TOPIC NO.22

37. The hydrolysis of starch by enzyme “amylase” produces maltose. In the human body maltose is further hydrolyzed by enzyme maltase (present in intestinal brush border) to produce….

a. Galactose
b. **glucose**
c. maltotrisoe
d. mannose

38. Maltose comprises of two glucose molecules that are joined together by….glycoside linkage.

a. **α(1-4)**
b. α(1-2)
c. β(1-2)
d. β(1-4)

TOPIC NO.23

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39. The source of milk that can be carried out by different microorganisms such as E.coli involves the conversion of lactose into lactic acid. This reaction is carried out by enzyme….

a. β Glucosidose
b. β lactam
c. **β golactosidase**
d. α lactose

40. all of following are true for lactose except;

a. **it is found abundantly in grape juice**
b. it is made up of galactose and glucose
c. it is a reducing sugar
d. it is dextrorotatory

TOPIC NO.24

41. The hydrolysis of sucrose to glucose and fructose is catalyzed by sucrose which is also present in intestinal brush border like lactose and maltose. Sucrose is also known as…..

a. Amylase
b. furanose
c. glycogen dehydrogenase
d. **invertase**

42. Sucrose commonly known as table sugar is formed by reaction between glucose (Glc) and fructose (Fru). The glycosidic linkage found in sucrose can be denoted as….

a. Fru (α1-2β) Glc
b. Glc (α1-4β) Fru
c. **Glc (α1-2β) Fru**
d. Fru (α1-6β) Glc

TOPIC NO.25

43. The human ABO blood groups best illustrate the biomedical importance of carbohydrates. The blood groups, A and B differ from O on the basis of antigen. Compared to O antigen, an extra monosaccharide is present in both A and B. For antigen A the extra monosaccharide molecule is:

a. **N-acetyl galactosamine**
b. ribose
c. **c.** glucose
d. N-acetyl muramic acid

TOPIC NO.27

44. Immunoglobulins are….in nature.

a. Carbohydrates
b. **proteins**
c. lipids
d. none of these

45. Proteins present in cell membrane may function as….or transporters.

a. Storage proteins
b. **receptors**
c. immunoglobulin
d. hormones

TOPIC NO.29

46. The protein…..formed by all vertebrates, is the chief structural component of hair, scales, horn, wool, nails and feathers.

a. Luciferin
b. heamoglobin
c. **c. keratin**
d. collagen

TOPIC NO.30

47. All of following are true for amino acids except:

a. The side chains of amino acids influence their solubility in water
b. The carbon atom present between amino group and carboxyl group in amino acid is known as a carbon
c. In formation of a protein amino acids are joined together by a linkage known as peptide bond.

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d. The genetic code specifies and codes for both standard and non-standard amino acids.

TOPIC NO.31

48. Which of following is found in brain tissue?

a. D-alanine and D-glutamate
c. **free D-serine and D-aspartate**
b. free D-serine and D-alanine
d. free D-serine and D-glutamate

49. D-amino acids are also non-standard amino acids that occur naturally. Which of following D-amino acid is found in cell wall of gram positive bacteria?

a. 5-hydroxy lysine
c. cysteine
b. **D-alanine**
d. D-serine

50. Some proteins contain additional amino acids that arise by modification of an amino acid already present in a peptide i.e. after the protein has been synthesized. An example of such an amino acid is:

a. **5-hydroxylysine**
c. lysine
b. peptidyle proline
d. glutamic acid

TOPIC NO.32

51. Apart from 20 standard amino acids, some other amino acids may also be synthesized and become a part of protein. An example of such an amino acid…..which is a naturally occurring, genetically coded amino acid used by some methanogenic archea.

a. Sialic acid
c. aspartic acid
b. **pyrrolysine**
d. phenylalanine

TOPIC NO.33

52. The pKa value for acetic acid is……

a. 3.76
c. 5.76
b. 7.66
d. **4.76**

53. The R group is hydrogen atom in…..

a. Alanine
c. proline
b. **glycine**
d. lysine

54. All amino acids except…..have a chiral carbon and have two possible isomers.

a. **Glycine**
c. glutamic acid
b. lysine
d. tryptophan

Week no.3

TOPIC NO.35

55. Which statement about amino acids at physiological PH is true?

a. There is no charge on either carboxyl group or amino group
b. The carboxyl group is dissociated (-coo-) and the amino group is protonated (NH_3^+)
c. Only carboxyl group is dissociated (-coo-)
d. Only amino group is protonated (NH_3^+)

Topic no.38

56. The net charge on alanine in acidic solution (PH less than 2) is…..

a. First negative then becomes neutral
c. neutral
b. negative
d. **positive**

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57. An amino acid that contains a single carboxyl group and a single amino group will have a net positive charge at acidic PH since both groups are….

a. **Protonated**
c. deprotonated
b. oxidized
d. hydrolyzed

TOPIC NO.40

58. The pK2 value for glycine is….

a. 7.65
c. **9.60**
b. 5.97
d. 2.34

59. The characteristic PH at which net electric charge on amino acid molecule is zero is called….

a. Ampholytic point
c. cationic PH
b. **isoelectric point**
d. anionic point

TOPIC NO.41

60. A buffer is a solution that resists change in PH following addition of and acid or base. Among amino acids only……has on R group (pKa=6.0) providing significant buffering power near neutral PH usually found in intracellular fluids of most animals.

a. Glycine
c. aspartic acid
b. glucose
d. **histidine**

61. The number of dissociable hydrogen in histidine is..,..

a. **Three**
c. two
b. zero
d. one

TOPIC NO.44

62. Based upon properties of R group, mainly polarity and ability to interact with water, amino acids may be classified into…. Main classes.

a. Eleven
c. two
b. **five**
d. three

63. Which statement about non-polar aliphatic amino acids is false?

a. They are lipid-like and promote hydrophobic interactions.
b. They have side chains that do not bind or give off protons.
c. The side chains often cluster towards the interior of protein.
d. They form hydrogen bonds abundantly.

TOPIC NO.45

64. Out of 20 standard amino acids……contain a secondary amino group and is called an imino acid.

a. Methionine
c. histidine
b. serine
d. **proline**

65. Which amino acid causes least steric hinderance and is often found in bends or tightly packed chains of fibrous protein?

a. Tyrosine
c. alanine
b. **glycine**
d. proline

66. Which of following is non-polar, aliphatic amino acid?

a. **Leucine**
c. phenylalanine
b. tyrosine
d. glutamine

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TOPIC NO.46

67. Which of following is positively charged at physiological PH?

a. **Arginine**
c. aspartate
b. glutamate
d. tyrosine

68. Among aromatic amino acids…..is non-polar while……is polar.

a. Tryptophan, lysine
c. **phenylalanine, tyrosine**
b. phenylalanine, glycine
d. tyrosine, alanine

TOPIC NO.47

69. The amide group of….serves as a site of attachment for oligosaccharide chains in glycoproteins.

a. Threonine
c. **asparagine**
b. serine
d. all of above

70. Aliphatic polar amino acids are…..in nature.

a. **Hydrophilic**
c. non-reactive
b. hydrophobic
d. aromatic ringed tryptophan

TOPIC NO.48

71. The side chain of….amino acids are proton acceptor

a. Acidic group
c. alpha carbon
b. negatively charged
d. **lysine**

TOPIC NO.51

72. Immunoglobulins are….proteins and have…..and…..components.

a. Enzymatic, variable, constant
c. protective, non-variable, constant
b. **protective, variable, constant**
d. catalytic, variable, constant

73. Immunoglobulin and…..are form of protective proteins.

a. Plasminogen
c. hemoglobin
b. **fibrinogen**
d. igG

74. Catalytic proteins are called as……

a. Carbon R chain groups
c. amino group proteins
b. catalysts
d. **enzymes**

75. The protein responsible for blood clotting is called:

a. Collagen
c. heamoglobin
b. **fibrinogen**
d. immunoglobulin

76. …..is an example of a contractile protein.

a. Heamoglobin
c. amylase
b. **myosin**
d. fibrinogen

Week no.4

TOPIC NO.52

77. Storage proteins have ability to bind and store specific elements or compounds…..is an example of a storage protein that stores copper.

a. **Ceruloplasmin**
c. heamoglobin
b. ferritin
d. urease

78. Proteins involved in respiration include heamoglobin, myoglobin and…..

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a. Collagen
b. **cytochromes**
c. fibrin
d. cellulases

TOPIC NO.53

79. Almost all peptides bonds in a protein are present in the….configuration.
a. Cis-trans
b. **trans**
c. cis
d. D

80. The….bond formed by covalently joining of two amino acids through aminde linkage
a. Carboxyl
b. **peptide**
c. sulphide
d. amide

TOPIC NO.54

81. Three dimensional arrangement of two or more polypeptides is called
a. Primary structure
b. secondary structure
c. tertiary structure
d. **quaternary structure**

82. ….structure of protein refers to particularly stable arrangement of amino acid residues giving rise to recurring structure patterns.
a. Primary structure
b. **secondary structure**
c. tertiary structure
d. quaternary structure

TOPIC NO.55

83. The two ends of the polypeptide chain are known as the….and….terminus.
a. D, L
b. **C, N**
c. R, S
d. α,β

TOPIC NO.56

84. The folding of…..contiguous segments of polypeptide form secondary structure of protein.
a. 20
b. **3-30**
c. 3-20
d. 3-300

85. Which of following statements about secondary structure of protein is false?
a. The simplest arrangement that can be assumed by polypeptide chain is called α helix.
b. **Within alpha helix every peptide bond is involved in ionic interactions, disulphide linkages and hydrogen bonds**
c. The most stable and common secondary structure include alpha helix and beta sheets.
d. The secondary structure describes local spatial arrangement of main-chain atoms.

TOPIC NO.57

86. Which of following is example of non-polar alpha helix side chain?
a. Sulphide
b. **leucine**
c. arginine
d. tryptophan

87. In secondary structure of protein….normally reflects properties of R-group as…….
a. Beta helix, aromatic
b. **alpha helix, non-polar**
c. beta pleated sheets, polar
d. secondary form of alpha helix, non –polar

TOPIC NO.58

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88. Alpha helices are found in very diverse proteins such as….which is a fibrous protein and….which is a globular protein.

a. Amylase, myoglobin
b. keratin, amylase
c. **keratin, myoglobin**
d. cellulose, keratin

89. Alpha helices are most common secondary structure found in protein. Nearly all…proteins contain alpha helices in their membrane spanning domains.

a. Transport
b. globular
c. **Trans membrane**
d. protective

TOPIC NO.59

90. Clusters of twisted strands of beta sheet form the core of many…..proteins.

a. Collagenous
b. structural
c. fibrous
d. **globular**

91. Beta sheets are type of regular secondary structure that maximized hydrogen bonding between peptide backbones. The sheet is described as….if the polypeptide strands run in same direction (as defined by their amino and carboxyl terminal)

a. **Parallel**
b. anti-parallel
c. helical
d. aligned

TOPIC NO.60

92. Which of following secondary structure can be classified as non-regular and does not have repeating elements?

a. Parallel beta sheets
b. anti-parallel beta sheets
c. **beta bonds**
d. alpha helix

93. Which of following statements about beta bonds is false?

a. **For formation of bond carbonyl group of first residue forms a peptide bond with amino-group hydrogen of fourth residue.**
b. They are short regions usually involving four successive amino acid residues.
c. They often connect strands of antiparallel beta sheets
d. They reverse direction of polypeptide chain, helping it form compact globular shape.

TOPIC NO.61

94. Motifs are also known as super secondary structure. An example of a motif found in protein is….

a. **β Barrel**
b. α helix
c. β turns
d. β sheet

95. A…..also called a super secondary structure is simply a recognizable folding pattern involving two or more elements of secondary structure and connection between them.

a. **Motif**
b. polypeptide
c. domain
d. β sheet

TOPIC NO.62

96. Which statement best describes domains found in protein?

a. It is non-regular secondary structure that does not have repeating elements.
b. It is simply a recognizable folding pattern involving two or more elements of secondary structure and connection between them.

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c. **It is section of protein structure sufficient to perform a particular chemical or physical task such as binding of substrate or other ligand.**
d. It is the folding pattern of secondary structural elements into three-dimensional conformation.

TOPIC NO.63

97. Disulphide bonds are formed between two molecules of amino acid cysteine. The reaction involves:
a. **Oxidation of sulfhydryl groups** c. phosphorylation of alpha carbon
b. methylation of alpha carbon d. reduction of sulfhydryl groups

98. All of following are involved in stabilizing three dimensional tertiary structure of proteins except:
a. Disulphide bridges c. **glycosidic bonds**
b. hydrogen bonds d. hydrophobic interactions

99. Hydrophobic interactions occur between non-polar uncharged side chains of amino acids and help stabilize its three dimensional tertiary structure. An example of an amino acid most commonly involved in hydrophobic interactions is:
a. Alanine **c. phenylalanine**
b. aspartic acid d. glycine

100. Electrostatic interactions help stabilize three dimensional tertiary structure of proteins. These interactions are formed as a result of ionic bonds formed between positive and negative charges amino acids. The positive charges are usually present on….amino acids.
a. Acidic c. neutral
b. **basic** d. all

TOPIC NO.64

101. Many proteins have multiple polypeptide subunits (from two to hundreds). A multi subunit protein is also referred to as a multi-mer. The repeating structural unit in such a multi-meric protein is called a…..
a. Motif c. **protomer**
b. amino acid d. monosaccharide

TOPIC NO.65

102. Proteins may exist in form of multi-subunits. An advantage of multi subunit structure is that different subunits can have different activities and cooperate in a common function. This is best understood by structure of enzyme….which exists as a multi-protein complex.
a. **Pyruvate dehydrogenase** c. ribozyme
b. hemoglobin d. ig A

TOPIC NO.66

103. The three dimensional structure of a protein must have an external surface appropriate for its environment. For example plasma proteins contain…..amino acids on surface to remain soluble in as aqueous environment.
a. Non-polar c. **polar**
b. neutral d. hydrophobic

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104. A protein is called a….protein if its amino acid composition and molecular conformation are unchanged from that found in natural states.

a. Functional
b. folded
c. **native**
d. natural

TOPIC NO.67

105. The native conformation of protein is dependent on its folding patterns. These folding patterns are directed by:

a. Hydraulics
b. function of protein
c. conformation of peptide bonds
d. **thermodynamics**

TOPIC NO.68

106. The three dimensional structural of native protein in its physiological state is one in which Gibbs free energy of whole system is….,

a. **Lowest**
b. highest
c. 100
d. positive

Week no.5

TOPIC NO.69

107. Protein folding is governed by thermodynamics. The folding process involves a decrease in randomness and thus a decrease in….

a. Stability
b. interactions formed
c. flexibility
d. **entropy**

TOPIC NO.70

108. To overcome kinetic barriers while forming native conformation of protein, cell make use of a group of proteins known as:

a. **Auxiliary protein**
b. plastids
c. cytochromes
d. keratins

109. As a protein folds and refolds while it is searching for its native low energy state it passes through many high-energy conformations that slow process. What are these high energy conformation called?

a. **Kinetic barriers**
b. domains
c. super-secondary structures
d. motifs

TOPIC NO.71

110. The enzyme that catalyzes the interchange, or shuffling of disulfide bonds until bonds of native conformation are formed is called:

a. Protease
b. protein lyase
c. **protein disulfide isomerase**
d. sulfide dehydrogenase

111. Unfolded proteins that thermodynamically trapped in a misfolded dead may be unfolded by another group of proteins known as:

a. **Chaperones**
b. immunoglobulins
c. cytochromes
d. reducatases

TOPIC NO.72

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112. misfolding of proteins may occur spontaneously or as a result of mutation…..is a disease that results from misfolding of protein in body.

a. Tuberculosis
b. influenza
c. hepatitis
d. **Parkinson`s disease**

TOPIC NO.73

113. A loss of three dimensional structures sufficient to cause loss of function of protein is called….

a. Misfloding
b. **denaturation**
c. catalysis
d. reactivation

TOPIC NO.74

114. Certain proteins that are denatured by heat, extremes of PH or denaturing reagents can regain their structure and biological activity if returned to conditions in which native conformation is stable. An example is the enzyme ribo-nuclease which is denatured in a….solution in the presence of reducing agent.

a. **Concentrated urea**
b. normal saline
c. buffer
d. serum

TOPIC NO.75

115. A molecule bound reversibly by a protein is called a…..

a. Chelator
b. **ligand**
c. lysozyme
d. sub-unit

TOPIC NO.76

116. Excess iron has significant affinity for specific organs, particularly.

a. the lungs, liver and endocrine glands
b. **the heart, liver and endocrine glands**
c. endocrine glands
d. Liver only

117. The quaternary structure of human hemoglobin is best described as a

a. Tetramer of four different subunits
b. Tetramer of identical subunits
c. **tetramer of two different subunits**
d. dimer of two myoglobin dimers.

118. Once a heme group is oxidized, what molecule is produces?

a. Myoglobin
b. hemoglobin
c. **methemoglobin**
d. hemoglobin c

119. Which of following statements about erythrocytes is correct?

a. They fight infection
b. they are produced in spleen
c. **they lack a nucleus**
d. they clot blood

120. The oxygen binding site in hemoglobin and myoglobin?

a. **Heme hroup**
b. Ca^{+}
c. globin group
d. Mg^{+}

121. Folic acid aids;

a. **In maturation of red blood cells and is also required for RNA synthesis**
b. In maturation of red blood cells and is also required for DNA synthesis
c. In maturation of red blood cells and destruction of RNA
d. In maturation of red blood cells and destruction of DNA

122. The production of marrow cells takes place within:

a. Kidney
c. liver

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b. **the bone marrow** **d.** spleen

123. Approximately how many hemoglobin molecules are there in each red blood cell?
a. 5 million c. 50 million
b. 30 million d. **300 million**

124. What happen to red blood cells if heam group were removed from hemoglobin?
a. Blood clot formation would be inhibited
b. Red blood cell would not be able to reproduce
c. white blood cells would not be able to reproduce
d. **red blood cells would not be able to bind oxygen**

125. Anaemia can be classified into how many categories?
a. **3** c. 4
b. 2 d. 5

126. Myoglobin is composed of a single polypeptide chain which has……
a. Four O_2 binding sites c. three O_2 binding sites
b. **one O_2 binding site** d. two O_2 binding sites

TOPIC NO.77

127. second degradation product of hemoglobin is green pigment called as:
a. Bilirubin
b. **Biliverdin**

128. One degradation product of hemoglobin is brown bile pigment called as:
a. **Bilirubin**
b. Biliverdin

129. Haemoglobulin the red pigment in blood consist of protein component and iron complex of derivative
a. Iron c. **prophyrin**
b. protein d. benzene ring

130. Hemoglobin has a structure
a. Tertiary c. secondary
b. **quaternary** d. primary

131. Hemoglobin consists of pairs of different proteins, designated as alpha and beta chains.
a. Five c. three
b. **two** **d.** four

132. How many amino acids in the alpha chain of hemoglobin
a. **141** c. 543
b. 144 d. 146

133. How many amino acids in the beta chain of hemoglobin
a. **146** c. 141
b. 140 d. 435

134. Disulfide bonds most aften stabilize native structure of
a. Intracellular proteins c. **extracellular proteins**
b. dimeric proteins d. multi-subunit proteins

135. The second degradation product of hemoglobulin is green pigment biliverdin the result of breakdown by oxidation occurs in…
a. Kidney c. spleen
b. **liver** **d.** heart

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136. One degradation product of hemoglobulin is brown pigment bilirubin. This step occur in

a. Kidney
b. liver
c. spleen
d. **bile**

137. Hemoglobin binds O2 molecules

a. One
b. **four**
c. two
d. five

TOPIC NO.78

138. Identify the amino acids containing non-polar, aliphatic R groups.

a. Lysine, arginine, histidine
b. **glycine, alanine, leucine**
c. serine, threonine, cysteine
d. phenlalnine, tyrosine and tryptophan

139. Hemoglobin forms an unstable reversible bond with oxygen. In oxygenated state it is called…

a. Hemoglobin
b. **oxyhemoglobin**

140. Which of following is an imino acid?

a. **Proline**
b. alanine
c. histidine
d. lysine

141. Which among following is a non-essential amino acid?

a. Threonine
b. lysine
c. histidine
d. **serine**

142. The iron atom of heme has…..coordination bonds

a. Seven
b. **six**
c. nine
d. eight

TOPIC NO.80

143. Oxygen stored in red muscle myoglobin is released during O_2 deprivation (e.g severe exercise) to be used in muscle mitochondria for…..

a. Aerobic synthesis of more O_2
b. **aerobic synthesis of ATP molecules**
c. anaerobic synthesis of ATP molecules
d. anaerobic synthesis of more CO_2

144. In case of severe exercise the….stored in red muscle myoglobin is released.

a. **Oxygen**
b. carbon
c. carbon dioxide
d. ozone

145. The protein or globin portion of myoglobin thus creates special microenvironment for…..that permits reversible binding of one oxygen molecule (oxygenation)

a. **Heme**
b. Ca
c. Mg
d. none of above

146. The surface of myoglobin is polar important for interacting with…..aqueous environment of cytosol.

a. Non-polar
b. inorganic
c. organic
d. **polar**

147. The molecular weight of hemoglobin is about

a. 23,650
b. **64,450**
c. 62,450
d. 78,450

148. The cervice (pocket) created by….amino acids in interior of myoglobin create a binding pocket for heme.

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a. Acidic
b. basic
c. polar
d. **non-polar**

149. The…..gas is used for aerobic synthesis of ATP in muscle mitochondria during case of severe exercise.

a. **Oxygen**
b. carbon dioxide
c. ozone
d. none

150. About….of amino acid residues in myoglobin are found in alpha helices with bends in between.

a. 68%
b. 10%
c. 58%
d. **78%**

151. The surface of myoglobin is…..important for interacting with polar aqueous environment of cytosol.

a. **Polar**
b. non-polar
c. aqueous
d. hard

152. Histidine is a generally considered to be a…..amino acid

a. Non-polar
b. **polar**

153. The example of natural porphyrins include…….

a. Myoglobin
b. haemoglobin
c. cytochrome C
d. **all of above**

154. Myoglobin is found in…….

a. **Heart and skeletal muscles**
b. pancreas
c. kidney
d. stomach

TOPIC NO.81

155. In myoglobin and hemoglobin heme is covalently linked with…..amino acid (eight residue of F helix)

a. Alanine FB
b. both
c. **histidine FB**
d. none of these

156. The proximal histidine FB of myoglobin binds directly to the….of heme group.

a. Zinc
b. magnesium
c. calcium
d. **iron**

TOPIC NO.83

157. the binding of oxygen to one heme group enables an oxygen binding to second heme group of some hemoglobin molecule is called…

a. structural binding
b. **cooperative binding**
c. amino acid binding
d. nucleic acid binding

158. The……change has no effect on function of myoglobin.

a. Amino acid
b. **Conformational**

159. The property of co-operative binding of oxygen of different subunits of macromolecule is applicable to….

a. Myoglobin
b. **Heamoglobin**

TOPIC NO.84

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160. A plot of degree of saturation (Y) measured at different partial pressures of oxygen (pO2) is called….
a. Carbon dissociation curve
b. **oxygen dissociation curve**
c. nitrogen dissociation curve
d. none of these

161. In contrast to myoglobin hemoglobin can bind…..oxygen molecules one at each of its heme group
a. Two
b. **four**
c. five
d. one

162. The conformational change of hemoglobin is usually described as changing from T (tense) state to R (relaxed) state with a low affinity for oxygen.
a. True
b. **false**

163. The partial pressure of oxygen needed to achieve half-saturation of binding sites is called…..
a. P60
b. P70
c. **P50**
d. none

164. During tense stage of hemoglobin binding sites are….
a. Closed
b. **hindered**
c. replaced
d. open

165. During relaxed stage of hemoglobin the binding sites are….
a. Closed
b. hindered
c. replaced
d. **opened**

166. The conformational change of hemoglobin is usually described as changing from…..state with low affinity for O2 to…..state with high affinity for O2
a. Tense to high tense
b. high tense to tense
c. **T (tense) to R (relaxed)**
d. R (relaxed) to T (tense)

167. The degree of saturation (Y) of oxygen-binding sites on all myoglobin or hemoglobin molecules can vary between…..and…..
a. Zero and 10 percent
b. 10 and 100%
c. 100%
d. **zero and 100%**

168. Myoglobin can bind only….molecule oxygen because it contain only one heme group
a. Three
b. four
c. **one**
d. two

169. During conformational change each successive addition of O2 shifts equilibrium further toward…..state thus addition of oxygen in lungs.
a. Relaxed or tense state
b. **relaxed state**
c. tense state
d. none of these

TOPIC NO.85

170. The partial pressure of oxygen (P50) is a conventional measure of hemoglobin affinity for…..
a. **Oxygen**
b. carbon
c. nitrogen
d. sulfur

171. The partial pressure of oxygen P50 is approximately……mm Hg for myoglobin
a. 40 mm Hg
b. 20 mm Hg
c. **1 mm Hg**
d. 15 mm Hg

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172. Degree of saturation (Y) (%) of oxygen binding sites on all myoglobin or hemoglobin molecules can be any value between.
a. -1 and +1
b. **0 and 100**
c. 0 and 1
d. none of these

173. Myoglobin can bind to one oxygen molecule because it contain only 1 heme group.
a. **True**
b. False

174. The plot of degree of saturation (Y) measured of different partial pressures of oxygen is called…..
a. Oxygen saturation curve
b. **oxygen dissociation curve**
c. calibration curve
d. oxygen partial pressure curve

175. …..is 1mmHg for myoglobin and 26.6 mmHg for hemoglobin
a. Total oxygen pressure
b. pulmonary oxygen pressure
c. **partial pressure of oxygen**
d. blood oxygen tension

176. Myoglobin can bind to four oxygen molecule because it contains 4 heme groups.
a. True
b. **False**

177. The partial pressure of oxygen is approximately….mmHg for hemoglobin.
a. 27.6 mmHg
b. **26.6 mmHg**
c. 25.7 mmHg
d. 24.5 mm Hg

Week no.6

TOPIC NO.86

178. The oxygen dissociation curve for myoglobin has….shape. This reflects the fact myoglobin reversibly binds a single molecule of oxygen
a. Longitudinal
b. **hyperbolic**
c. straight
d. transverse

179. During reversible binding of protein the oxygenated (MbO2) and deoxygenated (Mb) myoglobin exist in a….equilibrium. Mb + O2 = MbO2
a. Reverse
b. **simple**

180. The strenuous exercise lowers the PO2 of muscle tissue to about….mmHg hence permitting continued muscular activity
a. **5**
b. 4
c. 2
d. 3

TOPIC NO.87

181. The sigmoidal binding curve of hemoglobin for oxygen is possible due to….structure of hemoglobin.
a. **Multi-subunit**
b. single-subunit

182. Hemoglobin must bind oxygen efficiently in lungs where the PO2 is about….mmHg and release oxygen in tissue where PO2 is about….mmHg.
a. 60 and 35 to 40
b. 40 and 35 to 40
c. **100 and 35 to 40**
d. 50 and 35 to 40

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TOPIC NO.88

183. The net effect of affinity of hemoglobin for last oxygen bound is approximately….times greater than its affinity for first oxygen bound.

a. 365
c. 302
b. **300**
d. 301

TOPIC NO.92

184. The Bohr Effect was discovered by Danish physiologist who discovered it in 1904 named as….

a. Christian bohr
b. Bohr John

185. Hemoglobin transport about…..of total H+ and 15% to 20% of CO2 formed in tissues to lungs and kidneys.

a. 60%
c. **40%**
b. 70%
d. 50%

186. What is name of molecules formed when oxygen binds to hemoglobin?

a. **Oxyhemoglobin**
c. oxygen-hemoglobin
b. oxyglobin
d. none of these

187. This effect of PH and CO2 concentration on binding and release of oxygen by hemoglobin is called….

a. Nelson effect
c. nucleus effect
b. **Bohr effect**
d. electron removal

188. Fetal hemoglobin shifts oxygen saturation curve in what direction?

a. Right and down
b. left and down

189. The enzyme responsible for catalyzing conversion of water and carbon dioxide into carbonic acid is

a. Carbon dioxylase
c. carbonic hydrolase
b. **carbonic anhydrase**
d. carbonic hydroxylase

190. What are main things responsible for causing Bohr shift?

a. Increased carbon dioxide pressure
b. Carbon monoxide`s high affinity for hemoglobin
c. Increased oxygen requirements
d. increased H+ ion

191. Carbon dioxide and water combine to form

a. Hydrogen bicarbonate
c. hydrogen ions
b. **carbonic acid**
d. hydrogen carbonates ions

192. The ion found haem group is what….

a. O2-
c. Fe3+
b. Mg2+
d. **Fe2+**

193. At highest partial pressure of oxygen the saturation of oxygen slows down because

a. The hemoglobin is too cold
b. the hemoglobin becomes denatured
c. **the hemoglobin becomes saturation n**

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d. the oxygen is displaced by carbon dioxide

194. Carbonic acid dissociates to form

a. Hydrogen carbonate ions and hydrogen ions
c. H+
b. carbonic acid anhydrase
d. **H+ and H2CO3-**

195. The Bohr Effect/Shift moves oxygen curve in what direction

a. Down and right
b. down and left

196. When carbon dioxide binds directly with hemoglobin is forms

a. **Carboxyhemoglobin**
c. carbon dioxyhemoglobin
b. carbaminohemoglobin
d. carbglobin

197. During reversible of protein blood PH consequently raises the affinity of hemoglobin for oxygen increased and protein binds more O2 for transport to….

a. Lung
c. kidney
b. **peripheral tissues**
d. heart

198. At relatively low PH and high CO2 concentration of peripheral tissues affinity of hemoglobin for oxygen…..as H+ and CO2 are bound and O2 is released to tissues.

a. **Decrease**
b. incease

199. In lungs PH of blood is…..because CO2 is being exhaled.

a. Zero
c. **higher**
b. neutral
d. lower

TOPIC NO.93

200. The process by which oxygen enters blood from alveoli is

a. Diffusion
c. facilitated diffusion
b. **active transport**
d. none

201. Traditionally during reversible binding of protein curve starts with PH at…..temperature at…..centigrade and PaO2 at…..

a. 7.4, 42°C and 40mmHg
c. 8.4, 42°C and 50mmHg
b. 8.4, 42°C and 60mmHg
d. **7.4, 37°C and 40mmHg**

202. Once PaO2 reaches…..mmHg curve is almost flat indicating there is little change in saturation above this point.

a. 70
c. 90
b. 80
d. **60**

203. Oxyhemoglobin dissociation curve Bohr Effect decreased affinity of hemoglobin for….gas caused by increased of carbon dioxide PH etc.

a. Carbon dioxide
c. **oxygen**
b. nitrogen
d. ozone

TOPIC NO.100

204. The melting properties of fatty acids and lipid bilayers are due primarily to…..

a. **Vander Waals forces**
c. electrostatic interactions
b. hydrogen bonds
d. covalent bonds

205. Cholesterol is essential for normal membrane function because it…..

a. Catalyzes lipid flip-flop in bilayer
b. Cannot be made by higher organism e.g. mammals.
c. Spans the thickness of bilayer
d. **keep membrane fluid**

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206. The glycerophospholipids are composed of two fatty acids attached to glycerol and one of following head groups….
a. Choline phosphatase
b. serine dihydrase
c. ethanolamine and water
d. **threonine**

207. Phospholipids are molecules that contain…..
a. **hydrophilic head and hydrophobic tails**
b. positively charged functional groups
c. long water-soluble carbon chains
d. Cholesterol and water

208. Sickle cells are very fragile and rupture easily. This results in….
a. **Anemia**
b. hemophilia
c. aplastic anemia
d. blood cancer

209. Micelles of fatty acids in water are organized such that the….face solvent and the….are directed toward the interior.
a. Hydrophobic tails, hydrophilic heads
b. Hydrophobic heads, hydrophobic tails
c. **Carboxylic acid group, Hydrocarbon chains**
d. Hydrocarbon chains, carboxylic acid groups

210. Which of following fatty acids would have lowest critical micelle concentration…?
a. C6-COOH
b. C5-COOH
c. **C8-COOH**
d. C4-COOH

211. The main difference between saturated and unsaturated fatty acids is…
a. The presence of keto group
b. that one is absent from phospholipids
c. the number of carbons
d. **the presence of double bond**

TOPIC NO.101

212. Fats are abundantly found in
a. **Reproductive tissue**
b. Vegetative tissue

213. The example of phospholipid is….
a. Steroid
b. sterol
c. cholesterol
d. **lecithin**

214. All kind of lipids are….in water.
a. Miscible
b. soluble
c. **insoluble**
d. all of given

215. Natural lipids are readily soluble in.
a. Mercury
b. **organic solvent**
c. water
d. oil

216. …..are major structural elements of biological membranes.
a. **Phospholipids and sterols**
b. phospholipids
c. proteins
d. none of these

217. Fats are the….of fatty acids and glycerol.
a. Tautomer
b. isomer
c. ether
d. **esters**

218. Transporting lipids in blood are combination of lipids and….
a. Glycolipids
b. **proteins**
c. sphingolipids
d. glucose

219. Which of following occurs when hydrogen is reacted with vegetable oil?
a. The hydrogenated vegetable oil will become polarized.

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b. The hydrogenated vegetable oil will contain fewer tran`s` fats.
c. The hydrogenated vegetable oil will become solid at room temperature.
d. The hydrogenated vegetable all will become a saturated fat.

220. What are components of triglyceride molecule?
a. One glycerol and two fatty acid
b. One glycerol and two cholesterol
c. One glycerol and three fatty acid
d. One glycerol and one cholesterol

221. What is solubility of lipids in water?
a. Partially insoluble
b. partially soluble
c. insoluble
d. soluble

222. Which of following are saturated fatty acids?
a. Butyric acid
b. palmitic acid
c. linoleic acid
d. oleic acid

223. Amphipathic means that
a. One end of molecule carbohydrate, the other is protein
b. One end of molecule hydrophobic, the other is hydrophilic
c. One end of molecule negative, the other is positive
d. all of given

224. Which of following is characteristic of lipids?
a. They are composed of nitrogenous chains
b. They are polar molecules
c. they are either fats or oils.
d. they are not soluble in water

225. Name two essential fatty acids?
a. Lauric and myristic
b. linoleate and linolenate
c. arachidonic and oleic
d. oleic and linoleic

TOPIC NO.102

226. Lipids are classified in following three major groups:
a. Simple lipids
b. Precursor and derived lipids
c. complex lipids
d. all of above

227. Un-esterified fatty acids are transported in circulation in association with….
a. Albumin
b. gelatin
c. casein
d. none of these

228. Fatty acids are found in un-sterified form as….. , a transport form in plasma.
a. Boundary fatty acids
b. free fatty acids
c. oils
d. esters

229. Glycolipids (glycosphingolipids) contain….
a. Lipids containing sphingosine and carbohydrate
b. Lipids containing fatty acid, sphingosine and carbohydrate
c. Lipids containing fatty acid and carbohydrate
d. lipids containing fatty acids

230. In case of long-chain fatty acids (LCFA) the….protein is predominant.
a. Hydrophilic
b. Hydrophobic

231. Fatty acids occur in body mainly as……in natural fats and oils.
a. Protein
b. fatty acid
c. esters
d. oils

232. Esters of fatty acids with higher molecular weight monohydric alcohols (having one OH group).
a. Waxes
b. lipid
c. steroid
d. protein

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233. Low level of fatty acids occurs in all tissues, but substantial amounts sometimes can be found in the….. , particularly during fasting.
a. Stomach
b. blood
c. intestine
d. **plasma**

234. Fatty acids that occur in natural fats usually contain on even number of carbon atoms ranging from….to….carbons.
a. 2 to 38
b. 3 to 67
c. **4 to 36**
d. 3 to 37

235. Ester of fatty acids containing group in addition to an alcohol and fatty acid called as:
a. Simple lipid
b. derived lipid
c. **complex lipid**
d. none of these

236. This anionic group of fatty acid has an affinity for water, giving fatty acid it's…..nature.
a. **Amphipathic**
b. hydrophilic
c. polar
d. hydrophobic

237. derived lipids are composed of
a. hydrocarbon ring and long hydro carbon chain
b. ester of fatty acid and alcohol
c. ester of fatty acid
d. hydrocarbon ring

238. phospholipids frequently have nitrogen containing bases and other substituent are
a. glycerophospholipids the alcohol is glycerol
b. sphingophospholipids the alcohol is sphingosine
c. **both**
d. none of these

Week no.7

TOPIC NO.103

239. Fatty acids are classified on the basis of…
a. Degree of saturation
b. hydrocarbon chain length
c. dietary requirement
d. **all of above**

240. Fatty acids are classifies on the basis of hydrocarbon chain length include
a. Short chain fatty acids (2-4 C)
b. medium chain fatty acids (6-12 C)
c. long chain fatty acids (14-18 C)
d. **all of above**

TOPIC NO.104

241. Which is a characteristic of all fatty acid components in this lipid?
a. They all are joined to glycerol through an ester bond
b. **They all contain an un-branched carbon chain**
c. They all contain unconjugated cis double bonds
d. They all are hydrophilic because they contain oxygen

242. Which of following form of lipids are also referred as neutral lipids?
a. Steroid
b. **triacylglycerol**
c. wax
d. phospholipid

243. The melting point of fatty acids depends upon chain length and….
a. Charge of carbon
b. the position of double bond
c. **degree of unsaturation**
d. the shape of fatty acids

244. The following steroid present in cell membrane of fungi?

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a. Sitosterol c. stigmosterol
b. **ergosterol** d. campesterol

245. Which of following is a characteristic of both waxes and terpenes?
a. Both can contain fatty acid c. both can contain amino alcohol
b. both can be non-sponifiable d. **both can contain oxygen**

246. Which of following molecules is a typical fatty acid?
a. An amphipathic di-carboxylic acid with unconjugated double bonds
b. A molecule that has even number of carbon atoms in branched chain
c. A molecule that has one cis double bond in linear carbon chain
d. A polar hydrocarbon with that reacts with Noah to form a salt

247. Which of following is a characteristic of both triacylglycerols and glycerophospholipids
a. Both contain carboxyl groups and are amphipathic
b. Both contain glycerol and ether bonds
c. Both can be negatively charged at cellular PH
d. Both contain fatty acids and are saponifiable

248. Which characteristic does this lipid share with wax?
a. Both contain one or more carboxyl groups
b. Both contain three fatty acids
c. Both contain a polar head
d. Both contain one or more ester bonds

249. Which property does this lipid share with a typical triacylglycerol?
a. Both are amphipathic c. both contain long-chain alcohol
b. both contain on ether bond d. **both are saponifiable**

250. Fatty acids are carboxylic acids in which length of…..chains range from 4 to 38 carbons.
a. **Hydrocarbon** c. monosaccharide
b. polycarbon d. disaccharide

TOPIC NO.105

251. ……that us a fatty acid with one double bond and is abbreviated as 18:1
a. Glacial acetic acid c. tartaric acid
b. **oleic acid** d. palmitic acid

252. Which property can be shared by lipid and a terpene?
a. Both can contain isoprene
b. both can contain a saturated fatty acid
c. **Both can be very hydrophobic molecules** d. both can form micelles.

253. Which would be property of all major types of lipids in this membrane?
a. They would be saponifiable in base and hydrolyzed in acid
b. They would have polar head and non-polar tails
c. They would be joined to each other through covalent bonds
d. They would be composed of five carbon units

254. 18-carbon oleic acid with one double bond is abbreviated as….
a. 18-0 c. 18-1
b. 18:0 d. **18:1**

255. The 18-carbon with one double bond named as
a. **Oleic acid** c. butyric acid

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b. palmitic acid d. propionic acid

256. The 16-carbon saturated fatty acid names as

a. Oleic acid c. **palmitic acid**

b. propionic acid d. butyric acid

257. Based on its saturated similarity to other lipids, lipids most likely functions as….

a. A sex hormone c. **an energy storage molecule**

b. a membrane component d. vitamin required for vision

258. The most frequently used systematic nomenclature names the fatty acid after hydrocarbon with –oic being substituted for final……

a. **E** c. d

b. b d. a

259. which type of membrane lipid contain an acidic oligosaccharide

a. phosphatidylinositol c. cerebroside

b. globoside d. **ganglioside**

260. The carbon atoms in fatty acids are numbered with carboxyl carbon as…

a. C4 c. **C1**

b. C2 d. C3

261. Cis-9-hexadecenoic acid that is a fatty acid with nutritional significance is commonly known as…..acid

a. Oleic c. palmitic

b. **palmitoleic** d. none of given

262. The names of saturated fatty acids end in one of following suffixes.

a. -enoic c. -ol

b. **-anoic** d. –dehyde

263. Which membrane lipid contains an amide bond?

a. Phosphatidylserine c. cholesterol

b. **sphingomyelin** d. phosphatidic acid

264. Which is a characteristic of biological membranes?

a. Membranes contain proteins and amphipathic lipids.

b. Membranes contain lipids that polymerize into one large molecule

c. Membranes have hydrophobic groups on surfaces

d. Membranes have an asymmetrical micelle structure

265. The position of any…..in fatty acids are specified relative to carboxyl carbon by superscript numbers following Δ (delta)

a. **Double bonds** c. hydrogen atom

b. R-group d. carbon atom

266. The name of unsaturated acids with double bonds ends in…..

a. **Enoic** c. anoic

b. dnoic d. none of these

267. The name of saturated acids ends in….

a. **anoic** c. enoic

b. dnoic d. none

268. Eicosanoids are derived from either…..fatty acids.

a. Omega-6 c. omega-3

b. **omega-3 or omega-6** d. none of given

269. Which is characteristic of lipids in a biological membrane?

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a. The fluidity of membrane decrease with lower levels of saturated fatty acids
b. Lipid molecules are held in fixed position by non-covalent bonds with proteins
c. The fatty acids of lipid molecules are found in interior of membrane
d. Specific glycerophospholipids are distributed equally on two membrane surface

TOPIC NO.106

270. The family of polyunsaturated fatty acids (PUFAs) with double bond between then third and fourth carbon from methyl end of chain special importance in human…..
a. **Nutrition**
b. Mental health

271. In poly unsaturated fatty acid the carbon of methyl group is known as…..carbon
a. **Omega**
b. bravo
c. Mu
d. alpha

272. The solubility of fatty acids in water….
a. Increase with increase in chain length and few double bonds
b. Increase with increase in chain length and increase in double bond
c. Decrease with chain length and increase in double bond
d. none

273. the sterol act as precursor of vitamin D2
a. Protein
b. cholesterol
c. lanosterol
d. **ergosterol**

274. What are the functions of dietary fat?
a. Provide energy to body
b. form part of cellular membranes
c. cell signaling
d. **all of above**

275. The carbon of terminal methyl group is called….regardless of chain length.
a. Alpha
b. beta
c. **gamma**
d. none

TOPIC NO.107

276. Physical and chemical properties of fatty acids are due to their….
a. Side chain attached to all hydrogen atoms
b. Length of hydrocarbon chain and degree of saturation
c. Length of polypeptide chain and hydrocarbon atoms
d. Name of fatty acid

TOPIC NO.108

277. Hydrogenation of fatty acids leads to hardening of natural oils and formation of margarine because:
a. Cis double bonds are converted to Trans double bonds
b. Double bonds are converted to sign bonds
c. Tran's double bonds are converted to Cis double bonds
d. Single bonds are converted to double bonds

TOPIC NO.110

278. Hydrogenated fats are used by many food producers to provide following properties
a. Provide rich texture
c. increased shelf life

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b. increase melting point d. **all of above**

279. Fatty acids produce alkyl alcohols by…of carboxylic group.
a. Oxidation c. esterification
b. methylation d. **reduction**

280. The physical state of fat at room temperature is….
a. Liquid c. plasma
b. **solid** d. gas

281. Which of followings not lipid?
a. Oil c. fat
b. **protein** d. wax

282. In case of soap micelle the….hydrocarbon chain cluster in inside and….carboxylate group lie on surface
a. Polar, non-polar
b. Non-polar, polar

283. Partial hydrogenation has an undesirable effect including….
a. Cis bond convert to Trans
b. Trans bond convert to Cis

284. The following salt is water soluble
a. Ca c. Zn
b. Mg d. **K**

285. In human diet all cholesterol comes from…
a. Beans c. minerals
b. vegetable product d. **animal product**

286. During formation of detergent….reaction of carboxyl group of fatty acid results in production of alkyl alcohols.
a. Reduction
b. Oxidation

287. Lauric acid saturated fatty acid having….carbon atom chain
a. 14 c. **12**
b. 13 d. 15

288. Which of following is example of derived lipid…..?
a. Steroid c. carotenoid
b. terpenes d. **all of above**

289. Which of following protects our heart and kidneys from injury?
a. Lubricants c. skin
b. muscle d. **fat**

290. The process which convert unsaturated fatty acid to saturated fatty acid….
a. Glycolysis c. liquefaction
b. proteolysis d. **hydrogenation**

291. Margarines are vegetable oils treated with partial hydrogenation to form semi solid…
a. oil c. **butter**
b. fat d. protein

292. The following salt is water insoluble.
a. **Mg** c. Zn
b. K d. Na

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293. Margarines are vegetable….
a. Protein
b. wax
c. carbohydrate
d. **oil**

Finalterm

TOPIC NO.115

294. The first fatty acid produces during fatty acid synthesis in the body and precursor to longer fatty acids is…….
a. **Palmitic acid**
b. arachidonic acid
c. oleic acid
d. linoleic acid

week no.8

TOPIC NO.120

295. Glycerol combines with three molecules of…..to form glycerol nitrate.
a. Chloroform
b. methanol
c. **nitric acid**
d. iodine

296. Which of following compound is an example of a lipid?
a. Amino acid
b. water
c. **cholesterol**
d. nucleotides

297. Which of following give positive test for acrolein?
a. Cholesterol
b. sphingosine
c. cholesterol and sphingosine
d. **glycerol**

TOPIC NO.121

298. Following is an example of cell type specialized for storing large amounts of fats.
a. Heart cell
b. skin cells
c. erythrocytes
d. **adipocytes**

299. …….are the example of simplest lipids.
a. Sphingolipids
b. terpenoids
c. waxes
d. **triacylglycerols**

300. Simple lipids are ester of…..and alcohols
a. Water molecules
b. amino acids
c. glucose molecules
d. **fatty acids**

301. Waxes are an example of
a. Nucleotides
b. **lipids**
c. carbohydrates
d. protein

TOPIC NO.122

302. Triacylglycerol's are stored in the……of many plants as oil.
a. Root
b. leaves
c. **seeds**
d. stem

TOPIC NO.124

303. Triacylglycerols are insoluble in… …….
a. **Water**
b. Chloroform
c. Ether
d. Acetone

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TOPIC NO.125

304. Saponification number is the number of mgs of… …required to saponify the free and combine fatty acids in one gram of given fat:

a. HCL
b. **NAOH**
c. KCL
d. NACL

TOPIC NO.127

305. Linoleic acid is… …… carbon compound:

a. 6
b. 20
c. 10
d. **18**

306. Which one of following is not halogen

a. **Carbon**
b. Chlorine
c. Iodine
d. Bromine

TOPIC NO.128

307. Which one of following is used to prevent oxidative rancidity?

a. Air
b. Room temperature
c. **Vitamin E**
d. Air and room temperature

TOPIC NO.129

308. Cetyl alcohol was first isolated from…….

a. Bees
b. Plant cuticle
c. **Sperm whale**
d. Mollusks

TOPIC NO.130

309. Mericyl palmitate is made from:

a. Triacylglycerol and myristoleate
b. **Palmitic acid and mericyl alcohol**
c. Palmitic acid and cetyl alcohol
d. Glutamic acid and mericyl alcohol

TOPIC NO.132

310. In VLDL, the concentration of Triacylglycerol is:

a. 5%
b. **50-60%**
c. 90%
d. 10-12%

TOPIC NO.133

311. VLDLs are assembled in the ……..

a. Kidney
b. Lungs
c. **Liver**
d. Brain

TOPIC NO.150

312. Bile helps in… …….. digestion :

a. Iron
b. **Lipids**
c. Nucleotides
d. Amino acid

313. Bile is made and released by the ……..

a. Heart
b. Lungs
c. **Liver**
d. Brain

314. Primary bile acids are synthesized in the liver from… …..

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a. **Cholesterol** c. Glucose
b. Fructose d. Amino acid

TOPIC NO.153

315. Thromboxanes (tx) are formed in… …….
a. Glycolysis c. Kreb's cycle
b. Glycolytic pathway d. **Cyclooxygenase pathway**

TOPIC NO.154

316. Removal of glutamate residue from LTC4 forms… ……
a. **LTD4** c. LTD3
b. LTD2 d. LTD1

317. Leukotrienes were so named because they were first discovered in… …….
a. Skin cells c. Red blood cells
b. **White blood cells** d. None of above

318. When the oxygen is added, to the double bond it isomerizes to a position one carbon away from the Hydro-peroxy group and is transformed from the……configuration
a. trans to trans c. cis to cis
b. **cis to trans** d. trans to cis

319. The hydro-peroxy group is unstable and can be converted to the more stable… …….. group.
a. Hydroxy-eicosa c. **Hydroxyl**
b. Hydroxy-peroxy d. None

TOPIC NO.157

320. Nucleoside lipid derivatives such as CDP acyl glycerol are Intermediates in…
a. Sugar biosynthesis c. Protein biosynthesis
b. **Lipid biosynthesis** d. None of above

TOPIC NO.158

321. Energy dependent processes include:
a. Muscle contraction c. Motility of spermatozoa
b. Transport of nutrients across cell membrane d. **All of above**

322. Nucleoside tri-and diphosphate such as……..are principal donors and acceptors of phosphoryl group in metabolism.
a. ADP c. ATP
b. **Both** d. None

TOPIC NO.159

323. In living cells, Nucleotides and their derivatives can serve as:
a. Carriers of metabolic energy c. Intercellular signals
b. Enzyme co-factors d. **All of above**

324. Nucleotides also serve as carriers of activated intermediates in the synthesis of…
a. Proteins c. Lipids
b. Carbohydrates d. **All**

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TOPIC NO.160

325. When linked to vitamins nucleotides are structural components of several essential coenzymes, for example:

a. **Coenzyme A**
b. Coenzyme B
c. Coenzyme C
d. Coenzyme D

326. Which one is example of coenzyme:

a. NAD+
b. NAP+
c. FAD
d. **All**

TOPIC NO.161

327. Nucleotides are composed of:

a. Pentose monosaccharide
b. One, two, or three phosphate groups
c. A nitrogenous base (purines and pyrimidines)
d. **All**

TOPIC NO.163

328. The sugar molecule is present in nucleotide is

a. Triose
b. Tetrose
c. **Pentose**
d. Hexose

329. Which of following is true of the Pentoses found in nucleic acid?

a. The bond that joins nitrogenous bases to Pentoses is an O- glycosidic bond
b. The straight – chain and ring forms undergo constant interconversion
c. **The Pentoses are always in the β-furanose forms**
d. C-5 and C-1 of the pentose are joined to phosphate

TOPIC NO.164

330. ATP is a… ……

a. Nucleic acid
b. **Nucleotide**
c. Vitamin
d. Nucleoside

331. Nucleoside contains:

a. **Base-sugar**
b. Base-phosphate.
c. Sugar-phosphate
d. Base-sugar phosphate

TOPIC NO.165

332. Which of the following nucleotide contains only ribose sugar and not deoxyribose?

a. Thymine-Pentose sugar phosphate
b. Cytosine-pentose sugar phosphate
c. **Uracil-Pentose sugar phosphate**
d. Thymine-Pentose sugar phosphate

333. The difference between Thymine and Uracil is

a. One Hydroxyl group on the ribose ring
b. One methylene group on the pyrimidine ring
c. One amine group on the pyrimidine ring
d. **One methyl group on the pyrimidine ring**

TOPIC NO.166

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334. Compounds that generate nitrous acid (such as nitrites, nitrates, and nitrosamines) change the DNA molecules by… ……

a. Breakage of phosphodiester bond
c. Depuration
b. **Deamination of bases**
d. Formation of thymine dimer

335. Purines have… …….. rings (s),(each) containing… …..nitrogen (s), whereas pyrimidines have… …. Rings(s), (each) containing… ……. Nitrogen

a. 1;1;1;1
c. **2;1;1;2**
b. 2;2;1;2
d. 2;2;1;1

336. Purine base found in RNA is:

a. Thymine
c. Guanine
b. **Uracil**
d. Cytosine

TOPIC NO.167

337. Adenosine 3'-phosphate -5'-phosphosulphate is the… ……donor:

a. **Sulfate**
c. Phosphate
b. Adenosine
d. None

338. S-adenosylmethionine is a… ……group donor:

a. Sulfate
c. **Adenosine**
b. Methyl
d. Phosphate

339. When double-stranded DNA is heated at neutral pH, which change does not occur?

a. The hydrogen bonds between A and T breaks
b. **The covalent N-glycosidic bond between the base and the pentose** breaks
c. The absorption of ultraviolet (260nm)light increases
d. The helical structure unwinds

TOPIC NO.168

340. ……. play key roles in activating and inhibiting proteins in various cellular signaling cascade:

a. GDP
c. None
b. GTP
d. **Both**

341. The difference between a ribonucleotide and a deoxyribonucleotide is:

a. A ribonucleotide has an extra -OH at C-4
b. **A deoxyribonucleotide has an -H instead of an -OH at C-2**
c. A deoxyribonucleotide has an -H instead of an -OH at C-3
d. A ribonucleotide has more structural flexibility than deoxyribonucleotide

342. Find the correct statement about phosphodiester linkage between adjacent nucleotides in nucleic acids:

a. **5'-phosphate of one nucleotide joins the 3'-hydroxyl of the next nucleotide**
b. 5'-phosphate of one nucleotide joins the 5'-hydroxyl of the next nucleotide
c. 3'-phosphate of one nucleotide joins the 5'-hydroxyl of the next nucleotide
d. 3'-phosphate of one nucleotide joins the 3'-hydroxyl of the next nucleotide

TOPIC NO.169

343. The phosphodiester bonds that link adjacent nucleotides in both RNA and DNA

a. Always links A with T and G with C
b. are susceptible for alkaline hydrolysis
c. **Join the 3'Hydroxyl of one nucleotide to the 5'Hydroxyl of the next**

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d. From between the planar rings of the adjacent bases

344. Phosphorylation can… ….. ATP–dependent enzyme depending on the type of enzyme

a. Inhibit
c. None
b. Activate
d. **Both**

345. A phosphodiester bond is present in:

a. Fatty acid in a diglyceride
c. Amino acid in polypeptide
b. **Nucleic acid in a nucleotide**
d. Monosaccharide in a polysaccharide

TOPIC NO.170

346. Which of the following deoxyoligonucleotide will hybridize with a DNA containing the sequence:

a. **(5')GACCAGTCT(3')**
c. (5')CTCATTGAG(3)
b. (5')TCTGACCAG(3)
d. (5')GAGTCAACT(3')

347. The 'energy carrier ATP' is an example of a (n):

a. Deoxyribonucleotide triphosphate
c. Di-nucleotide
b. Ribonucleotide
d. **Ribonucleoside triphosphate**

TOPIC NO.171

348. By convention, the structure of a single strand of nucleic acid is always written with the

a. 3' end at the left and the 5' end at the right
b. 5' end at the left and the 3' end at the right
c. **5' end at the right and the 3'end at the left**
d. none of the above

TOPIC NO.172

349. Deoxyribose phosphate backbone is… …..and thus it is on the outside of the molecule.

a. stacked inside the molecule
c. hydrophobic
b. **hydrophilic**
d. none

350. Which one of following is a palindromic sequence

a. AGGTCC TCCAGG
c. **CCTTCC GCAAGG**
b. GGATCC CCTAGG
d. GAATCC CTTAGG

351. DNA replication is……..

a. Conservative
c. **Non-conservative**
b. Semi-conservative
d. None

352. In double-stranded nucleic acid, cytosine typically base pairs with:

a. Inosine
c. Thymine
b. **Guanine**
d. Adenosine

353. DNA model was presented Watson and Crick in… ……..

a. **1953**
c. 1951
b. 1957
d. 1955

354. Certain anticancer drugs exert their cytotoxic effect by intercalating into the…… groove of the DNA double helix, thus interfering with RNA and DNA synthesis:

a. Bases
c. **Narrow**
b. Phosphate backbone
d. Major

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355. These…… provide access for the binding of regulatory proteins to their specific recognition sequences along the DNA:
a. phosphodiester bond
c. **none of the above**
b. grooves
d. primary structure

356. In comparison with DNA-DNA double helices, the stability of DNA-RNA and RNA-RNA helices is:
a. DNA-DNA > RNA-RNA > DNA-RNA
b. RNA-RNA > DNA-RNA > DNA-DNA
c. RNA-RNA > DNA-DNA > DNA-RNA
d. RNA-DNA > RNA-RNA > DNA-DNA.

357. The bases in secondary structure of DNA are hydrophobic and are thus stacked inside the molecule.
a. outside of the molecule
c. **Hydrophobic**
b. hydrophilic
d. all of the above

358. Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid can be referred to as:
a. primary structure
c. tertiary structure
b. **Secondary structure**
d. All of above

TOPIC NO.173

359. The DNA oligonucleotide abbreviated PATCGAC
a. Has a phosphate on its 3' end
c. has seven phosphate groups.
b. **Has a hydroxyl at its 3' end**
d. has an A at its 3' end.

360. The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered……
a. Primary structure
c. **Secondary structure**
b. Tertiary structure
d. all

361. Which one is not a type of histone protein:
a. H3, H4
c. H2A and H2B
b. H1
d. **H6, H3A and H4B**

TOPIC NO.174

362. Z-DNA has a ……..
a. Uracil base
c. **Zig-Zag appearance**
b. Double helical nature
d. single stranded nature

363. The… ……. Strand is called template strand:
a. 5’-3
c. **3’-5**
b. 5’-5
d. 3'-3

364. DNA that is not copied during the synthesis of mRNA is called as
a. coding strand
c. mRNA-like strand
b. Non Template
d. **All of above**

TOPIC NO.175

365. According to Chargaff’s Rules, which one is not correct
a. Guanine equals the amount of cytosine
b. Guanine equals to the amount of thymine
c. Adenine equals the amount of thymine

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d. total amount of purines equals the total amount of pyrimidines

366. Chromatin is composed of

a. Nucleic acid only
c. Only protein
b. **Nucleic acid and protein**
d. None of these

367. These----· bonds, plus the van der Waals and hydrophobic interactions between the adjacent stacked bases, stabilize the structure of the double helix.

a. Phosphodiester
c. **Hydrogen**
b. Nitrogen
d. None of them

TOPIC NO.177

368. Which of the following is not true of all naturally occuring DNA?

a. The amount of A always equals the amount of T
b. **The ratio A+T/G+C is constant for all natural DNAS**
c. Deoxyribose units are connected by 3',5'-phosphodiester bonds
d. The two complementary strands are anti-parallel

TOPIC NO.178

369. Disruption of DNA can occur in the laboratory with alternation in

a. Heated above 80C
c. Salt concentration
b. PH
d. **All of above**

370. Separation of the two strands of the double helix when…….between the paired bases are disrupted.

a. Phosphodiester
c. **Hydrogen**
b. glycosidic bond
d. None

371. At neutral pH denaturation of a double helical RNA often requires temperatures ---- C or higher than those required for denaturation of a DNA molecule with comparable sequence.

a. 200
c. 100
b. 50
d. **20**

372. RNA duplexes are…….. Stable than DNA duplexes

a. Equal
c. **More**
b. Less
d. None of these

TOPIC NO.179

373. Denaturation can be monitored by measuring its absorbance at ------ nm.

a. **260**
c. 280
b. 300
d. 230

374. The tm of DNA is influenced by… ……..

a. base composition of the DNA
b. higher the content of GC base pairs
c. salt concentration of the solution
d. **all of above**

375. Higher GC base pairs increase Tm, because, with three hydrogen bonds, require- ---heat energy to dissociate than AT base pairs.

a. **More**
c. Less
b. Equal
d. None

376. Single-stranded DNA has a- relative absorbance at 260nm wavelength than does double strand DNA.

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a. Lower
b. **Higher**
c. Equal
d. None

TOPIC NO.180

377. Under appropriate condition DNA will form a hybrid with complementary
a. DNA
b. **Both**
c. RNA
d. None

378. Hybridization… ….is combined with gel electrophoresis techniques that separate nucleic acids by
a. Charge
b. **size**
c. Colour
d. Base composition

379. Under appropriate conditions (temp. & salt concentration), separated strands of DNA will renature or reassociate and form the double helix by the process called-----
a. renaturation
b. Reannealing
c. Hybridization
d. **all**

TOPIC NO.181

380. Uridine present in RNA is
a. Pyrimidine
b. nucleotides
c. **nucleoside**
d. purine

381. The function of RNAs (tRNA)
a. Contribute to the formation and function of ribosomes
b. adapter molecules that carry specific amino acids for protein synthesis
c. Contribute to the formation and function of the ribosomes
d. All of these

382. In which organelle, RNA is not found
a. Nucleolus
b. mitochondria
c. ribosomes
d. **cytoskeleton**

383. Some RNA molecules have intrinsic catalytic activity these RNA enzymes, are Called……..
a. proteases
b. small nuclear RNA
c. **ribozymes**
d. RNases

TOPIC NO.182

384. RNA can be hydrolyzed by alkali to 2',3' cyclic diesters of the mononucleotides, compounds that cannot be formed from alkali-treated DNA because of the absence of a----
a. **2'-hydroxyl group**
b. none
c. phosphodiester bond
d. deoxy ribose sugar

385. In RNA, the sugar moiety to which the phosphates and purine and pyrimidine bases are attached is
a. 5'-deoxyribose
b. 3'-deoxyribose
c. **Ribose**
d. 2'-deoxyribose

386. Pyrimidine: The pyrimidine components of RNA differ from those of DNA Instead of thymine, RNA contains the ribonucleotide of
a. Guanine
b. cytisine
c. adenine
d. **uracil**

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387. Chargaff's Rules Do Not Apply on RNA, so which one is not true for RNA:

a. G#C
b. Purines # Pyrimidines
c. A#U
d. **T#U**

TOPIC NO.183

388. Which of the following are possible base compositions for single-stranded RNA? %A %G %C %T %U

a. 25 25 25 0 25
b. 35 10 30 0 25
c. 5 45 45 0 5
d. **All of the above**

TOPIC NO.184

389. Double – stranded regions of RNA:

a. **can form between two self-complementary regions of the same single strand of RNA**
b. can be observed in the laboratory, but probably have no biological relevanc
c. Less stable than double-stranded regions of DNA
d. Do not occur

SUBJECTIVE

1. Properties of Nitrogenous Bases

- Aromatic: The Nitrogen containing bases are aromatic i.e. they have alternate double bonds
- Heterocyclic: They are heterocyclic i.e. structures that contain other atoms in addition to carbon, such as nitrogen in the ring structure. The six-atom rings of purines and pyrimidines are numbered in opposite directions.
- Weak Bases: Purines or pyrimidines with an –NH2 group are weak bases
- Functional Groups: The most important functional groups of pyrimidines and purines are ring nitrogen carbonyl groups exocyclic amino groups
- Hydrophobicity: The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral cell pH
- Stacking Interaction: Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids.
- Base stacking helps to minimize contact of the bases with water, and these interactions are very important in stabilizing the three-dimensional structure of nucleic acids.

2. Primary structure of DNA

The primary structure of a nucleic acid is its covalent structure and nucleotide sequence. The back bone of the primary structure is the linear strand made by sugar phosphate residues, linked together, while the bases project laterally. This way a long, un-branched chain is formed. Primary structure is a huge linear polymer of dNTPs that are joined to each other by 5'- 3' PDE bonds. The resulting long, un-branched chain has polarity. Both 5'-end and 3'-end are free. at 5'-end there is a free phosphate. at 3'-end there is a free OH that are not attached to other nucleotides. Purines and pyrimidines project laterally from the backbone and form a variable part.

- The variable part is concerned with the expression of genetic information.

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- By convention, the structure of a single strand of nucleic acid is always written with the 5′ end at the left and the 3′ end at the right • that is, in the 5′ to 3′ direction.
- Some simpler representations of this penta deoxy ribonucleotide are • pA-C-G-T-A OH, • pApCpGpTpA, and finally 5′-ACGTA-3′.

3. Function of Transfer RNA (tRNA)

- t RNA is the smallest of the three major species of RNA (4S).
- They are single stranded globular molecules.
- They remain largely in cytoplasm.
- They are generated by nuclear processing of a precursor molecule.
- tRNAs compose roughly 20% of total cellular RNA
- There are at least 20 species of tRNA molecules in every cell.
- Although each specific tRNA differs from the others in its sequence of nucleotides, the tRNA molecules as a class have many features in common
- Primary structure
- The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, and uracil but also contain unusual bases
- These unusual bases (also called modified bases) include Dihydrouracil, Pseudouridine and Thymine.
- t RNA is folded extensively.
- Extensive intra chain base pairing which leads to a characteristic CLOVER-LEAF structureo9
- T-RNA molecules consist of 74-95 nucleotides in a particular sequence.

4. What are simple lipids? Two types

Simple lipids
Esters of fatty acids with various alcohols
These contain: Fats (and Oils) and Waxes.
Fats: Esters of fatty acids with glycerol (Oils are fats in the liquid state)
Waxes: Esters of fatty acids with higher molecular weight monohydric alcohols.
(having one OH group)
Triacylglycerols (TAGs)
The simplest lipids constructed from fatty acids are the triacylglycerols,
Also referred to as; triglycerides, fats, or neutral fats or storage lipids.

5. Properties of waxes

- Waxes are insoluble in water, but
- Soluble in fat solvents and are negative for acrolein test. Very resistant to rancidity.
- Waxes are not easily hydrolyzed as the fats and are indigestible by lipases (enzymes responsible for fat digestion in body) •
- Thus they are of no nutritional value
- Waxes are of two types: True waxes • Other Waxes or Non true waxes or Wax-like compounds
- True Waxes are solid simple lipids containing a monohydric alcohol (with a higher molecular weight than glycerol esterified to long-chain fatty acids.

6. Reaction order with substrate concentration

Order of Reaction

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When [S] is much less, then the velocity of the reaction is approximately proportional to the substrate concentration.
The rate of reaction is then said to be first order with respect to substrate.
When [S] is much greater than Km the velocity is constant and equal to Vmax.
The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration

7. Difference between uracil and thymine

Thymine	Uracil
Thymine (T)—only in DNA	Uracil (U) —only in RNA
Thymine becomes thymidine and deoxythymidine	Uracil (U) becomes uridine and deoxyuridine depending on the type of sugar.
Thymine is 2, 4-dioxy-5-methyl-pyrimidine	Uracil is 2, 4-dioxy-pyrimidine
Methyl group present	Methyl group absent

8. What do you about kinetic enzyme?

The study of chemical reactions that are catalyzed by enzymes, in enzyme kinetics, we study about the reaction rate measure and the effect of change of condition.

9. What is the concentration of H in 0.1M NaOH?

Solution:
Kw= $[H^+][OH^-]$
With [OH-] = 0.1 M, solving for [H+] gives
$[H^+] = Kw/[OH^-]$
= 1 × 10-14M2/ 0.1M
= 10-14M2/0.1M
= 10-13M

10.Examples of hydrolases enzymes?

Common examples of hydrolases are:
Protein hydrolyzing Enzymes (peptidases)
Carbohydrates
Lipid hydrolyzing enzymes e.g. Lipases and Phospholipases

11.Define nucleotide?

Nucleotides are the building blocks of nucleic acids. Without them, DNA or RNA cannot be produced. Nucleotides are helpful in transmission of genetic information, in the synthesis of proteins and also act as energy currency in the cell

12.Describe induced fit hypothesis?

- Some proteins can change their shape (conformation)
- When a substrate combines with an enzyme, it induces a change in the enzyme's conformation
- This change in conformation when the substrate binds is induced by multiple weak interactions with the substrate.
- There may also be rearrangements of covalent bonds during an enzyme-catalyzed reaction.
- This conformational change is referred to as induced fit.

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13.Function of cAMP?

- Acts as second messenger in the cell
- It has role in glycogen metabolism
- cAMP, glycogenolysis
- cAMP TAG metabolism
- cAMP lipolysis
- It decreases cholesterol synthesis
- It causes activation of protein kinases which in turn; activate or deactivate other enzymes.
- It regulates the cell membrane permeability, by increasing permeability of cell membrane to H2O, Na +, K+& Ca +2
- Moreover, it regulates insulin secretion, catecholamine biosynthesis & Melatonin synthesis

14.Coenzyme

Some enzymes require no chemical groups for activity other than their amino acid residues.
Whereas some enzymes require molecules other than proteins for enzymic activity. If it is a complex organic molecule or metallo-organic compound it is termed a coenzyme.

15.Nucleotide as intermediate carrier?

- Carriers of intermediates: Nucleotides also serve as carriers of activated intermediates in the synthesis of some carbohydrates, lipids, and proteins.
- The sugar derivatives UDP-glucose and UDP-galactose participate in sugar inter conversions.
- And in the biosynthesis of starch and glycogen
- Similarly, nucleoside-lipid derivatives such as CDP acylglycerol are intermediates in lipid biosynthesis.

16.DNA denaturation

Separation of the two strands of the double helix when hydrogen bonds between the paired bases are disrupted.

17.Difference between temple and non-temple DNA

Template and Non-template Strands
The term template strand refers to the sequence of DNA that is copied during the synthesis of mRNA.
The opposite strand is called the Non Template or coding strand or the mRNA-like strand

18.Function of cGMP?

Cyclic GMP is synthesized from GTP
It serves as a second messenger in response to nitric oxide during relaxation of smooth muscle (especially blood vessels) so it has role in smooth muscle relaxation and vasodilatation. It also has role in Protein phosphorylation, Neurotransmission, Insulin action, Regulation of sodium channels

19.Mochils equation?

Leonor Michaelis and Maud Menten in 1913, proposed a simple model that accounts for most of the features of enzyme-catalyzed reactions. They postulated that the enzyme first combines reversibly with its substrate to form an enzyme-substrate complex in a relatively fast reversible step: k1

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E + S ↔ ES K-1
The ES complex then breaks down in a slower second step to yield the free enzyme (E) and the reaction product (P):
K2
ES ↔ E + P
K-2
Early in the reaction, the concentration of the product, [P], is negligible, and we make the simplifying assumption that the reverse reaction, P → S (described by k-2), can be ignored.
This assumption is not critical but it simplifies our task. The overall reaction then reduces to
k1 K2
E + S ↔ ES → E + P K-I
k1
E + S ES—k2 →E + P k-1
E = Enzyme S = Substrate P = Product ES = Enzyme-Substrate complex k1 rate constant for the forward reaction
K-1 = rate constant for the breakdown of the ES to substrate
k2 = rate constant for the formation of the products

20.Effect of platelets PAF

Platelets activating factor
This is an unusual ether glycero phospholipid, with a saturated alkyl group in an ether link to carbon 1 and an acetyl residue (rather than a fatty acid) at carbon 2 of the glycerol backbone
PAF is synthesized and released by a variety of cell types
It binds to surface receptors, triggering potent thrombotic and acute inflammatory events
It causes platelets to aggregate and degranulate (required for clotting), and neutrophils and alveolar macrophages to generate superoxide radicals (required for microbial killing)

21.Effect of PH in ionization of active site

The concentration of H+ affects reaction velocity in several ways.
First, the catalytic process usually requires that the enzyme and substrate have specific chemical groups in either an ionized or un-ionized state in order to interact.
For example, catalytic activity may require that an amino group of the enzyme be in the protonated form (–NH3 +).
At alkaline pH, this group is deprotonated, and the rate of the reaction, therefore, declines. Extremes of pH can also lead to denaturation of the enzyme.

22.Lanoline

Lanoline or wool fat is secreted by sheep sebaceous gland, and contain both free and esterified cholesterol e.g. cholesterol-palmitate
Lanoline secretion helps sheep in reducing water evaporation from skin. It is used as industrial lubricant and in cosmetics.

23.Iodine number

The number of grams of iodine which will be absorbed by 100 grams of a fat is termed its iodine number.

24.Small and large km

Small Km: A numerically small (low) Km reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is 1⁄2 Vmax

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Large Km: A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme

25.Hydrogenation

It is the addition of Hydrogen at the double bonds of unsaturated fatty acids. Hydrogenation converts unsaturated fatty acids to saturated fatty acids.

26.Cofactor

Some enzymes require no chemical groups for activity other than their amino acid residues. Whereas some enzymes require molecules other than proteins for enzymic activity. If the non-protein moiety is a metal ion such as Zn2+ or Fe2+, it is called a cofactor.

27.Activation energy

Energy required to start a chemical reaction is known as activation energy

28.Characteristics of palmitic acid

Palmitic Acid (16:0) Palmitic acid, or hexadecanoic acid, is the most common saturated Fatty Acid found in animals, plants and microorganisms • Palmitic acid mainly occurs as its ester in triglycerides (fats), especially palm oil. It is also found in high amounts in Butter, Cheese, milk and meat

Excess carbohydrates in the body are converted to palmitic acid. Palmitic acid is the first fatty acid produced during fatty acid synthesis and the precursor to longer fatty acids. As a consequence, palmitic acid is a major body component of fats found in the animals.

29.Pathway for metabolism of arachdonic acid

Omega-6 fatty acids

Linoleic acid (found in Vegetable oils, Safflower oil (LA) gamma Linolenic acid (GLA) Dihomo- -Linolenic acid (DGLA) Arachidonic acid (AA) (found in Meat Poultry Eggs)

30.Nucleotide as energy currency or cofactor

Energy currency: Nucleotides play an important role as "energy currency" in the cell.

Nucleoside tri- and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism.

By doing this, they play a key role in the energy transduction. This energy is used in almost energy requiring process in the body, such as; Muscle contraction, Transmission of nerve impulse, Transports of nutrients across cell membrane Motility of spermatozoa And many more energy dependent processes

31.Pka value of acetic acid

4.76

32.Steric acid has carbon number?

18

33.Types of RNA

There is a wide variety of RNAs,

- **Messenger RNAs (mRNAs)-** transfer genetic information from DNA to the proteinsynthesizing machinery.
- **Ribosomal RNAs (rRNAs)-** contribute to the formation and function of ribosomes
- **Transfer RNAs (tRNAs)-** adapter molecules that carry specific amino acids for protein synthesis
- **Small nuclear RNA (snRNA)-** play pivotal roles in RNA processing, particularly mRNA processing

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- **Ribozymes —** some RNA molecules have intrinsic catalytic activity these RNA enzymes, are called ribozyme

34.Nucleotide composition

Nucleotides are composed of

- A nitrogenous base (purine or pyrimidine)
- A pentose monosaccharide
- One, two, or three phosphate groups
- Nitrogenous Bases
- The nitrogen-containing bases belong to two families of compounds:
- Purines
- Pyrimidines

35.Rancidity of factors

The chemical deterioration of fats

When lipid-rich foods are exposed too long to the oxygen in air, they may spoil and become foul smelling. It occurs particularly on aging after exposure to atmospheric oxygen, light, moisture, bacterial or fungal contamination and/or heat. Saturated fats resist rancidity more than unsaturated fats that have unsaturated double bonds.

Rancidity is due to

Oxidation, Hydrolysis and Oxidative Rancidity

36.Examples of cyclic

There are two important cyclic nucleotides:

Cyclic AMP cAMP

Cyclic GMP cGMP

37.How DNA intact duplex?

DNA Renaturation

Under appropriate conditions (temp & salt concentration), separated strands of DNA will renature or reassociate and form the double helix by the process called renaturation (or reannealing). This reannealing process is also referred to as hybridization. When the temperature or pH is returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind, or anneal, to yield the intact duplex

38.Three functional group of nucleotide

Functional Groups: The most important functional groups of pyrimidines and purines are ring nitrogens, carbonyl groups, exocyclic amino groups

39.Subgroup of oxyreductase

Oxidoreductases

Catalyze oxidation reduction reactions further divided into four subgroups; Oxidase, Dehydrogenases, Hydroperoxidases and Oxygenases.

40.VLDL

VLDLs are assembled in the liver. Composed predominantly of TAGs synthesized in liver and contain some cholesterol and cholesteryl esters. As VLDL pass through the circulation, TAG is degraded and taken up by peripheral tissues in the form of fatty acids, causing the VLDL to decrease in size and become denser, called VLDL remnant.

41.Sterol

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Importantly, lipids provide the hydrophobic barrier that permits partitioning the aqueous contents of cells and subcellular structures as; phospholipids and sterols are the major structural elements of biological membranes.

42.Xanthine oxidase

Proteases hydrolyze proteins
Modifiers may precede the name to indicate, for example, the substrate (xanthine oxidase), the source of the enzyme (pancreatic ribonuclease), and its regulation (hormone-sensitive lipase)
Where needed, alphanumeric designators are added to identify multiple forms of an enzyme e.g, RNA polymerase III protein kinase C

43.Define rate of reaction

The reaction rate or rate of reaction is the speed at which a chemical reaction takes place, defined as proportional to the increase in the concentration of a product per unit time and to the decrease in the concentration of a reactant per unit time.

44.Hexokinase

Hexokinase catalyses the first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate. Hexokinase I, the isozyme in red blood cells has a low Km for glucose of approximately 0.05 mM- helpful in utilizing blood glucose even when the blood glucose concentration is very low.

45.Pka value of lactic acid

Calculate the pKa of lactic acid, given that when the concentration of lactic acid is 0.01M and the concentration of lactate is 0.087 M, the pH is 4.80
pH= pKa + log [lactate]/[lactic acid]
pKa = pH- log [lactate]/[lactic acid]
= 4.80 - log 0.087/0.01
= 4.80 - log 8.7
= 4.80 - 0.94
= 3.9

46.Function of lipoxin

A series of reductions of the resultant hydro-per-oxy groups leads to the formation of tri-hydroxy derivatives of arachidonic acid known as the lipoxins. Lipoxins induce chemotaxis and stimulate superoxide radicals for killing of microorganisms. Prostaglandins, thromboxanes, leukotrienes and lipoxins have very short half-lives and rapidly degraded in the body.

47.Cofactor and coenzymes difference

Coenzymes are small, non-protein organic molecules that carry chemical groups between enzymes (e.g. NAD and FAD). Forms easily removed loose bonds. Cofactor is a non-protein chemical compound that tightly and loosely binds with an enzyme or other protein molecules

48.Isomer of fructose D

The naturally occurring form of fructose is the D (–) isomer. In solution, glucose is dextrorotatory, and glucose solutions are sometimes known as dextrose

49.Interaction of DNA

Two basic types of chemical interaction with DNA are noncovalent and covalent binding. Covalent interactions occur when the chemical, or a portion of the chemical, is covalently adducted to the DNA helix. Proteins organize the DNA into a compact structure called chromatin. In eukaryotes, this structure involves DNA binding to a complex of small basic proteins called histones.

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50.sphingomylein

Present in the plasma membranes of animal cells and are especially prominent in nerve tissue including myelin, -thus the name "sphingomyelins. Sphingomyelin of the myelin sheath contains predominant longer-chain fatty acids such as lignoceric acid and nervonic acid (24 carbon) whereas gray matter of the brain has sphingomyelin that contains primarily stearic acid(18 carbon)

51.Hydrophobic and polar head

Sphingolipids, like other membrane lipids, are composed of a hydrophobic portion, (ceramide) and a polar head group

52.At high altitude PO2 is lower and oxygen to tissue is reduced

At high altitude of hills and mountains, where the pO2 is considerably lower (due to low atmospheric pressure). The delivery of oxygen to the tissues is now reduced. However, after just a few hours at the higher altitude, the BPG concentration in the blood begins to rise, leading to a decrease in the affinity of hemoglobin for oxygen. This adjustment in the BPG level has only a small effect on the binding of oxygen in the lungs. But a considerable effect on the release of oxygen in the tissues shown by increased oxygen unloading at the peripheral tissues. Increase in BPG concentration in the RBC shifts the dissociation curve to the right, i.e. increased O2 unloading. BPG concentration also increases in those conditions in which there is decreased O2 delivery to tissues, such as in anemia and respiratory diseases. As a result, the delivery of oxygen to the tissues is restored to nearly 40% of the oxygen that can be transported by the blood.

53.Enzyme is charted through rate of reaction

Relationship of velocity to enzyme concentration
The rate of the reaction is directly proportional to the enzyme concentration.
There is a linear relationship between reaction rate and enzyme concentration (at constant substrate concentration)

54.Phosphodiester bond

Phosphodiester Bond
When two or more nucleotides combine together a phosphodiester bond is formed.
This bond is formed mainly between the 3′OH group of sugar of one nucleotide and • 5′PO4 group of sugar of another nucleotide creating a phosphodiester linkage.

55.Acrolein test of glycerol

On heating with sulfuric acid or KHSO4 (dehydration)
It gives acrolein that has a bad odor. Used for detection of free glycerol or any compound containing glycerol. In contrast to glycerol Sphingosine does not show positive acrolein test.
Therefore glycerolipids and shingolipids can be differentiated on the basis of acrolein test.

56.Bile salt and its function

Bile salts (conjugated bile acids) are quantitatively the most important organic components of bile. The primary bile acids are synthesized in the liver from cholesterol. These are cholic acid and • chenodeoxycholic acid

Function of bile salt

The emulsification of dietary fats in intestinal canal, brought about by bile salts, is a prerequisite for digestion and absorption of fats. The bile salts, act to break apart the fat globules in the small intestines and allow them to become more "soluble" for absorption.

57.Activation energy relation to reaction

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The rate of a reaction reflects this activation energy: higher activation energy corresponds to a slower reaction. Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier. Alternatively, the activation energy can be lowered by adding a catalyst. Catalysts enhance reaction rates by lowering activation energies. The activation energy is lower when the enzyme catalyzes the reaction. The role of enzymes is to accelerate the inter-conversion of S and P. i.e enzymes lower the energy of activation, Δ G‡, of a reaction.

58.Hetrogenous mRNA

Messenger RNA (mRNA)

- ♥ This class is the most heterogeneous in
- ♥ Abundance • Size (500-6000 nucleotides) base sequence
- ♥ Stability
- ♥ mRNA comprise about 2–5% of total cellular RNA
- ♥ mRNA molecules are formed with the help of DNA template strand (3′-5′) during the process called transcription.
- ♥ In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated,
- ♥ there are untranslated regions at its 5′ and 3′ ends
- ♥ Moreover, there is a 5′ cap and a poly A tail at 3′ end

59.Sphingolipids

Sphingosine is an amino alcohol, which is a component of the class of lipids known as sphingolipids • Sphingosine is synthesized in the body in the form of ceramide, to which different moieties are added to form sphingolipids.

60.Histone and its classes

Eukaryotic DNA is associated with tightly bound basic proteins, called histones.
These serve to order the DNA into fundamental structural units, called nucleosomes.
There are five classes of histones, designated H1, H2A, H2B, H3, and H4.
These are positively charged at physiologic pH as a result of their high content of lysine and arginine. Two molecules each of H2A, H2B, H3, and H4 form the structural core of the nucleosome. Around this core, a segment of the DNA double helix is wound nearly twiceapproximately 140bp

61.Calculate ratio of concentration of acetic acid if PH= 5.40 and pka is 4.45

pH = pKa + log [acetate]/[acetic acid] • log [acetate]/[acetic acid]= pH- pKa
= 5.30-4.76
= 0.54
[acetate] / [acetic acid] = antilog 0.54
= 3.5
In summary,
When [HA] = [A-] ; pH=pKa
When [HA] > [A-] ; pH<pKa
When [HA] < [A-]; pH>pKa

62.Three nucleotides

Nucleotides are composed of three subunit molecules: a nucleobase, a five-carbon sugar (ribose or deoxyribose), and a phosphate group consisting of one to three phosphates. three nucleotides

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called a triplet or codon—codes for one particular amino acid in the protein. The nucleotide sequence in the DNA is first transcribed into a molecule of messenger RNA (ribonucleic acid).

63.Physical properties of fats

Neutral fats are
Colorless, odorless and tasteless substances •
Solubility: They are insoluble in water but soluble in organic fat solvents (e.g., ether, benzene, acetone, chloroform)
Specific gravity: The specific gravity of all fats is less than 1.0, consequently all fats float in water
Emulsification: Emulsions of fat may be made by shaking vigorously in water and by emulsifying agents such as gums and soaps. The emulsification of dietary fats in intestinal canal, brought about by bile salts, is a prerequisite for digestion and absorption of fats. The bile salts, act to break apart the fat globules in the small intestines and allow them to become more "soluble" for absorption. The hydrophobic fat molecules will clump together into globules in the watery mixture in the digestive system. The emulsifiers break them down to smaller "globules" and allow them to become more soluble.

64.What are lyses

An enzyme which catalysis the joining of specified molecules or groups by a double bond

65.Define saponification

Saponification
Hydrolysis of a fat by an alkali is called saponification
The resultant products are; glycerol and the alkali salts of the fatty acids, which are called “soaps”. The number of mgs of NaOH/KOH required to saponify the free and combined FA in one gram of a given fat is called its saponification number

66.Name the inorganic elements serve as cofactor

Metal ion such as Zn2+ or Fe2+,

67.Which amino acid form Disulphide Bridge

The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

68.Enzyme Kinetics

Enzymes are protein catalysts that, like all catalysts, speed up the rate of a chemical reaction without being used up in the process.
Enzyme kinetics is the study of the chemical reactions that are catalysed by enzymes. In enzyme kinetics, the reaction rate is measured and the effects of varying the conditions of the reaction are investigated. Studying an enzyme's kinetics in this way can reveal the catalytic mechanism of this enzyme, its role in metabolism, how its activity is controlled, and how a drug or an agonist might inhibit the enzyme. Enzyme kinetics is the investigation of how subsrtate bind with enzyme them

69.Hydrogenation of fats

Hydrogenation converts liquid vegetable oils into solid or semi-solid fats, such as those present in margarine. Changing the degree of saturation of the fat changes some important physical properties, such as the melting range, which is why liquid oils become semi- solid. presence of a catalyst such as nickel, palladium or platinum.

70.N glycosidic bond

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Aglycosidic bond orglycosidic linkage isatypeofcovalentbond that joins carbohydrate molecule to another group, which may or may not be another carbohydrate

71.Write the composition of triacyglcerols

A triglyceride is a lipid molecule made up of one unit Of **glycerol** and three **fatty acids**, hence the tri- prefix, which means three. A triglyceride looks a little bit like a creature with three tails. The head is **glycerol**, which is a simple sugar alcohol compound.

72.Write two examples of cyclic nucelotides

A cyclic nucleotide (cNMP) is a single-phosphate nucleotide with a cyclic bond arrangement between the sugar and phosphate groups. Like other nucleotides cyclic nucleotides are composed of three functional groups: a sugar, a nitrogenous base, and a single phosphate group. The two most well-studies cyclic nucleotides are cuclis AMP (Campp) and cyclic GMP (cGMP) while cyclic CMP (cCMP) and **cyclic** UMP (cUMP) are less understood.

Hydrolytic rancidity refers to the odor that develops when triglycerides are hydrolyzed and free fatty acids are released. This reaction of lipid with water may require a catalyst, leading to the formation of free fatty acids and glycerol. In particular, short-chain fatty acids, such as butyric acid, are malodorous

73.Halogenation

Similar to hydrogenation,

Halogens such as chlorine, bromine and iodine can also be added to double bonds in unsaturated fatty acids.

It is a very important property to determine the degree of unsaturation of the fat or oil that determines its biological value.

The degree of unsaturation is reflected by Iodine number. Iodine number is defined as the number of grams of iodine absorbed by 100gm of fat.

Fats rich in saturated fatty acids have low iodine numbers, while fats rich in unsaturated fatty acids have high iodine numbers

The determination of iodine number is useful to the chemist in determining the quality of oil or its freedom from adulteration

Iodine number of cotton seed oil varies from 103 to 111. That of olive oil from 79 to 88,

And that of linseed oil from 175 to 202. A commercial lot of olive oil which has iodine number higher than 88 might havebeen adulterated with cotton seed oil. The higher is the iodine number, the more reactive, less stable, more susceptible to oxidation and rancidification is the oil or fat.

74.What are enzymes? Write its classification with example

A several complex protein that are produced by cell and act as catalysts by specific biochemicl reaction. There were six classes of **enzymes** that were created so that **enzymes** could easily be named.These classes are: Oxidoreductases,Transferases,Hydrolases,Lyases, Isomerases, and Ligases. This is the international **classification** used for **enzymes**.

75.Spermaceti

Spermaceti is a wax that is most often found in the head cavities of the sperm whale. It was used in cosmetics, pharmacy and also in candles. Recent international regulation concerning whale captures has stopped its use.

76.Lipoprotein

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Lipoprotein is substance made of protein and fat that carry cholesterol through your bloodstream. There are two main types of cholesterol: high-density lipoprotein or good cholesterol. Low-density lipoprotein or bad cholesterol

77.Examples of coenzyme

FAD (Flavin Adenine Dinucleotide), NAD+ (Nicotin amidea denine Dinucleotide) and NADP+ (Nicotinamide adenine Dinucleotide Phosphate)

Eample:Acetic acid such as sodium acetate (CH3COOH) in which they have conjugate acid . Ammonia (NH3) and a salt(Nacl) in which they have conjugate base.

78.Properties of glycerol

Glycerol is a colorless, odorless liquid with sweet taste. Glycerol is seen in biological system as an carbohydrate and lipid metabolism because surplus carbohydrate can be converted into long chain fatty acids

79.Buffer define and its composition

A buffer is a solution containing either a weak acid or its salt or a weak base and its salt, which is resistant to change in PH. Buffer, are used to maintain a stable PH in solution, as they can neutralize small quantities of additional of base

- A weak base and a conjugate acid
- A weak acid and conjugate base
- The use of one or more depends upon the desired pH when preparing the buffer.

80.Nucleic acid medical application

Nucleic acid testing helps to identify genetic variations and predicts predisposition to cancer, alter diagnostic categories, enhance treatment strategies, enables early detection and prevention and improve outcomes for cancer patients the nucleic acids, DNA or RNA— for diagnosis or to provide guidance on therapy. as anti-viral drugs such as in the treatment of AIDS.

81.Two properties of glycerol trinitrate

Glycerol combines with three molecules of nitric acid to form Glycerol trinitrate that is used as explosive and vasodilator (*Vasodilators* are medications that open (dilate) blood vessels)

82.Differentiate between oxidative and hydrolytic rancidity

Oxidative rancidity is a natural process that affects fats and oil.

Hydrolytic rancidity refers to the odor that develops when triglycerides are hydrolyzed and free fatty acids are released. This reaction of lipid with water may require a catalyst, leading to The formation of free fatty acids and glycerol. In particular, short-chain fatty acids, such as butyric acid, are malodorous

83.Write hydrogenation process with respect to fats

84. During hydrogenation, vegetable oils are reacted with hydrogen gas at about 60ºC. A nickel catalyst is used to speed up the reaction. The double bonds are converted to single bonds in the reaction. In this way unsaturated fats can be made into saturated fats – they are hardened.

85.Role of nucleotides as coenzyme and intermediate carrier

ATP, an adenine nucleotide, is a universal energy currency in the cells of biological systems. Adenine nucleotides are components of three major coenzymes, NAD+, FAD, and CoA, organic molecules that assist in various biochemical reactions by serving as carriers. Nucleotides also function as regulators of metabolism.

86.Iodine number

- Iodine number is defined as the number of grams of iodine absorbed by 100 gm off at.

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- ♥ The more the iodine number, the greater the degree of unsaturation.
- ♥ Fats rich in saturated fatty acids have low iodine numbers,
- ♥ while fats rich in unsaturated fatty acids have high iodine numbers

87.cGMA

Cyclic guanosine monophosphate (cGMP) is a cyclic nucleotide derived from guanosine triphosphate (GTP). cGMP acts as a second messenger much like cyclic AMP. cGMA is a common regulator of tissue. In blood vessels relaxation of vascular smooth muscles lead to vascodilation and increased blood flow

88.carbon number in pentose sugar of nucleotides and nycleosides

Numbering of Carbon and Nitrogen Atoms

The carbon and nitrogen atoms in the rings of the base and the sugar arenumbered separately
The atoms in the rings of the bases are numbered
1 to 6 in pyrimidines &
1 to 9 in purines

In the pentoses of nucleotides and nucleosides the carbon numbers are given aprime (') designation to distinguish them from the numbered atoms of the nitrogenousbase.

The carbons in the pentose are numbered 1' to 5 '. Numerals with a prime (e.g., 2' or 3') distinguish atoms of the sugar from those ofthe heterocycle. when the 5'-carbon of a nucleoside (or nucleotide) is referred to, a carbon atom in the pentose, rather than an atom in the base, is being specified.

89.Nucleoside vs. nucleotide

A nucleoside consists of a nitrogenous base covalently attached to a sugar (ribose or deoxyribose) but without the phosphate group. A
Nucleotide consists of a nitrogenous base, a sugar (ribose or deoxyribose) and one to three phosphate groups.
Nucleoside = Sugar + Base
Nucleotide = Sugar + Base + Phosphate

90.How light produces in fireflies

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase.
RBC contains large amounts of the oxygen-transporting protein hemoglobin. (c) The protein keratin, formed by all vertebrates, is the chief structural component of hair, scales, horn, wool, nails, and feathers.

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